Roma Tre

INGEGNERIA
CIVILE, INFORMATICA E DELLE TECNOLOGIE AERONAUTICHE

# Information Visualization

## Graph Drawing

*G. Di Battista, M. Patrignani*

 copyright ©2025 g.di battista, m.patrignani

# Copyright notice

- All the pages/slides in this presentation, including but not limited to, images, photos, animations, videos, sounds, music, and text (hereby referred to as “material”) are protected by copyright
- This material, with the exception of some multimedia elements licensed by other organizations, is property of the authors and/or organizations appearing in the first slide
- This material, or its parts, can be reproduced and used for didactical purposes within universities and schools, provided that this happens for non-profit purposes
- Any other use is prohibited, unless explicitly authorized by the authors on the basis of an explicit agreement
- This copyright notice must always be redistributed together with the material, or its portions

 copyright ©2025 g.di battista, m.patrignani

# Overview

- Why graphs and what is graph drawing?
- Drawing conventions and aesthetics
- Decomposition of a graph
- A divide-and-conquer algorithm for deciding if a graph is planar

 copyright ©2025 g.di battista, m.patrignani

# Binary relationships

- One of the most recurring relationships
- A binary relationship is naturally represented by a graph

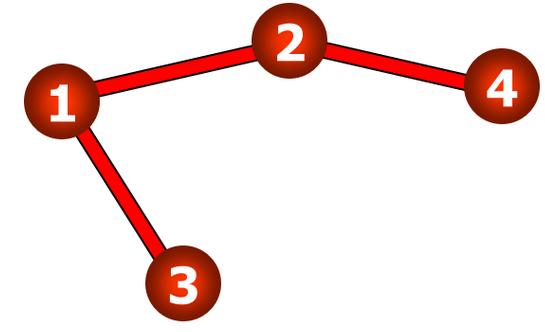
People

| | ID | | | | |
|---|---|---|---|---|---|
| | 1 | | | | |
| | 2 | | | | |
| | 3 | | | | |
| | 4 | | | | |

Friends

| | ID | ID |
|---|---|---|
| | 1 | 2 |
| | 1 | 3 |
| | 2 | 4 |

A dataset




A possible representation of the dataset

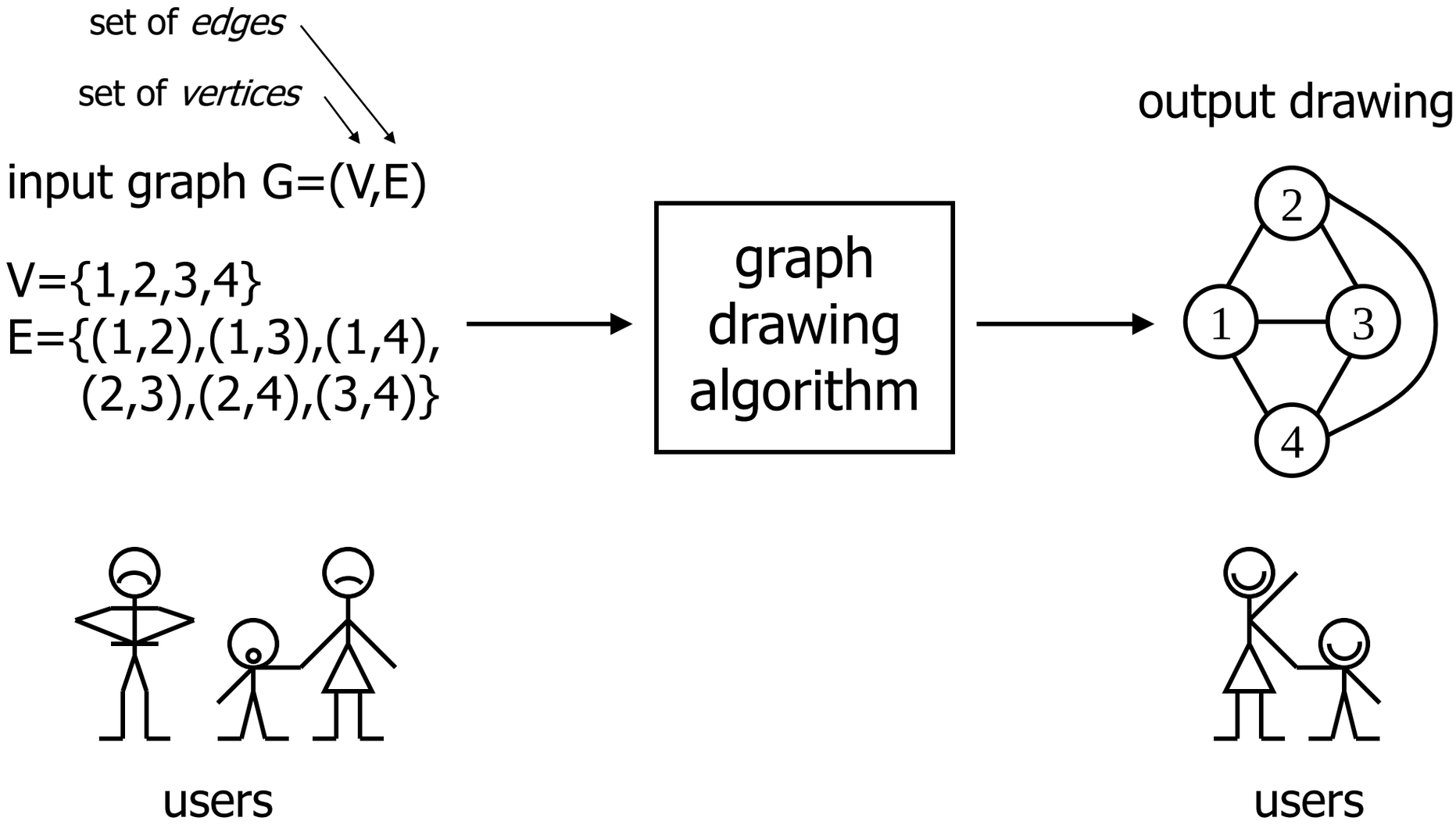
 copyright ©2025 g.di battista, m.patrignani

# Graph drawing

“most appealing subject in algorithms research”
[Donald E. Knuth, 1996]

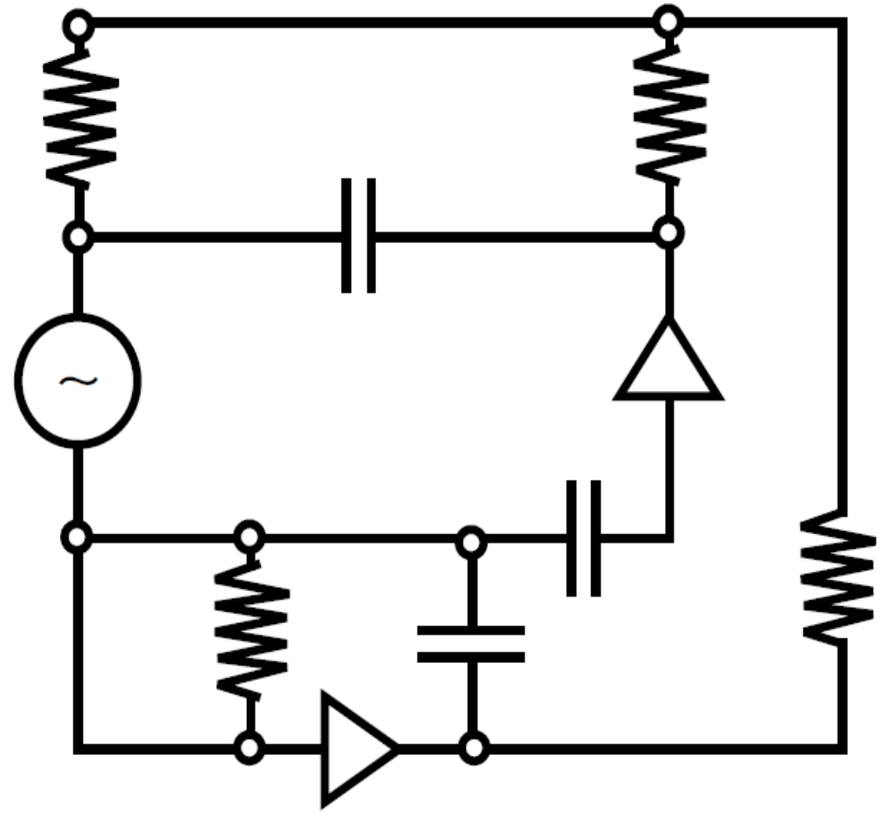
 copyright ©2025 g.di battista, m.patrignani

# Graph drawing purpuses



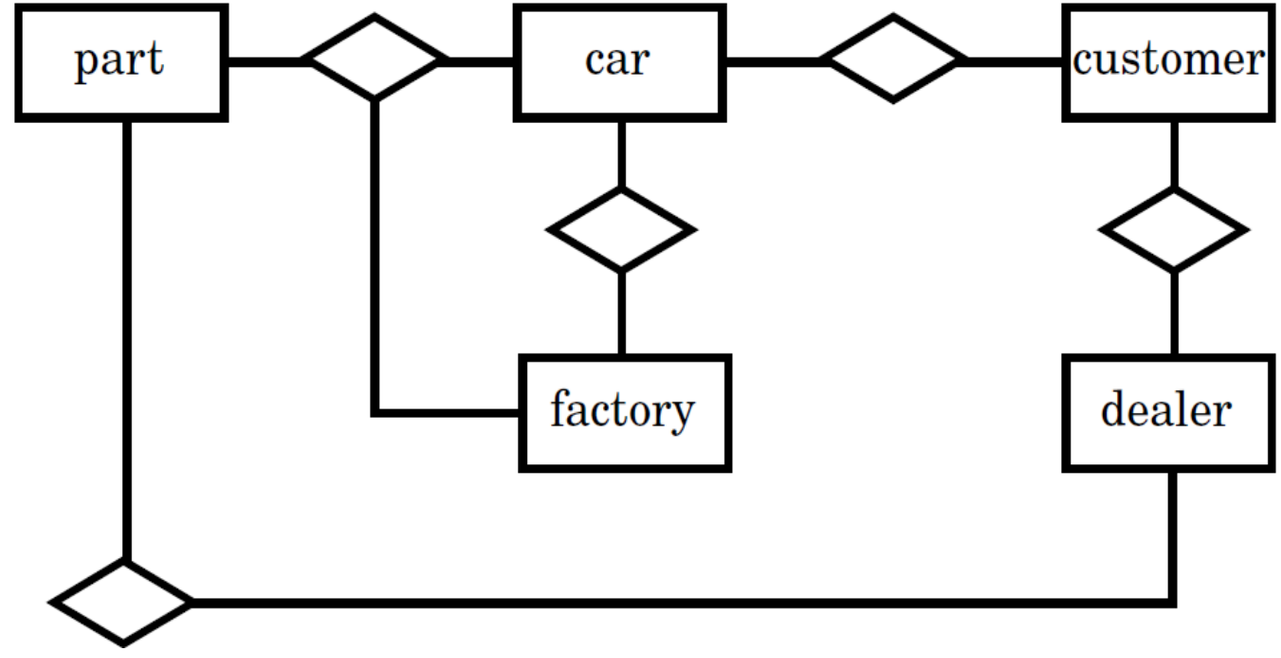

 copyright ©2025 g.di battista, m.patrignani

# Applications: electric circuits



 copyright ©2025 g.di battista, m.patrignani

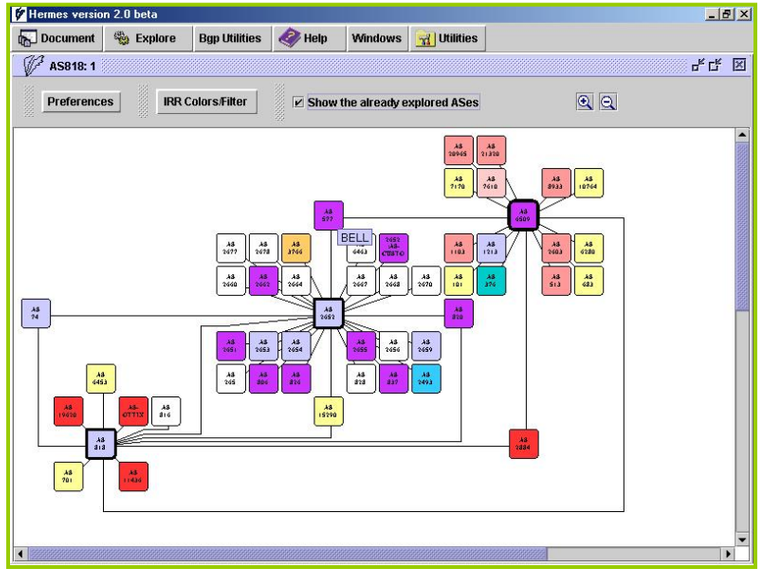
# Applications: engineering schemes

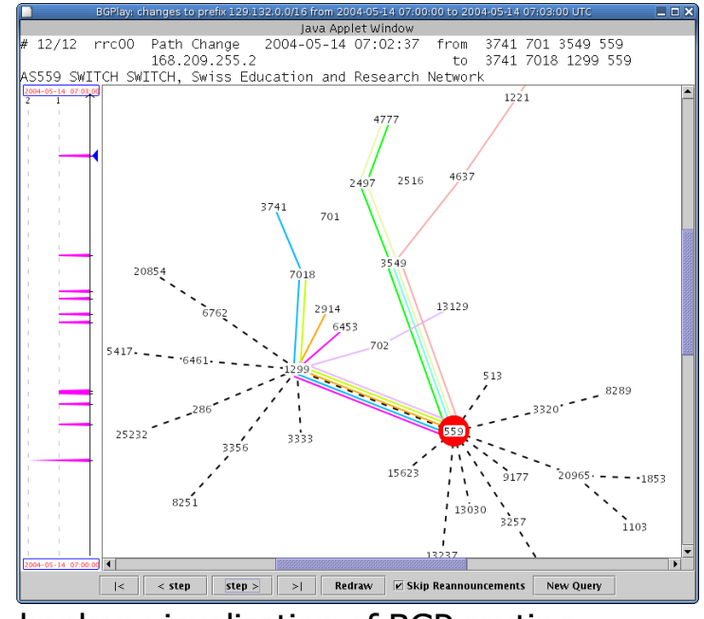



 copyright ©2025 g.di battista, m.patrignani

# Applications: computer networks

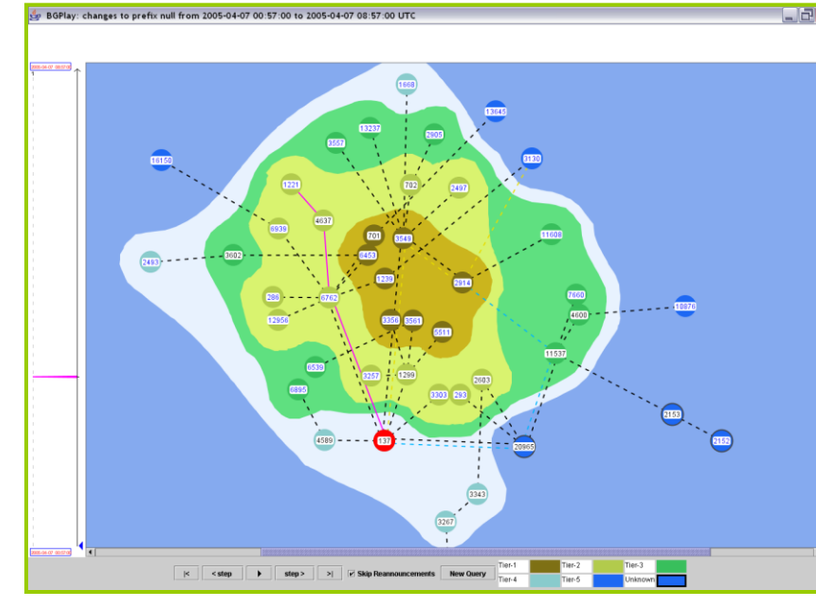
hermes: visualization of AS relationships




bgplay: visualization of BGP routing

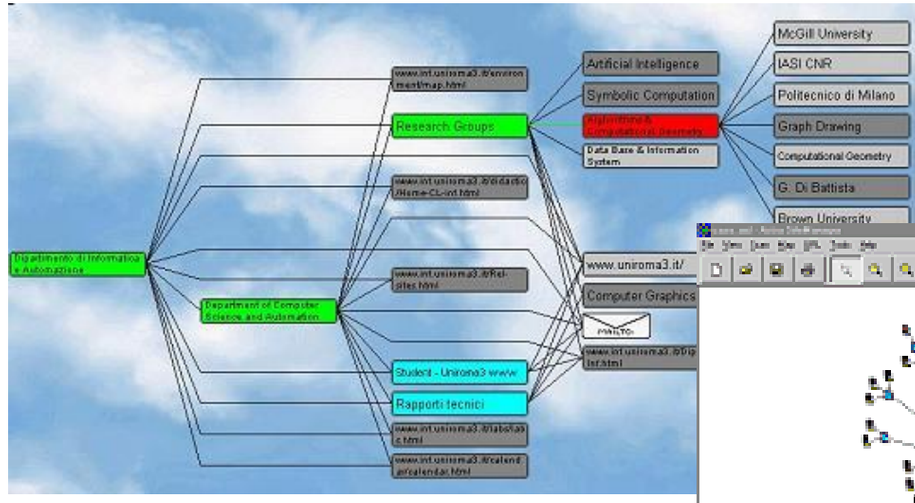
bgplay-island: visualization of AS hierarchy



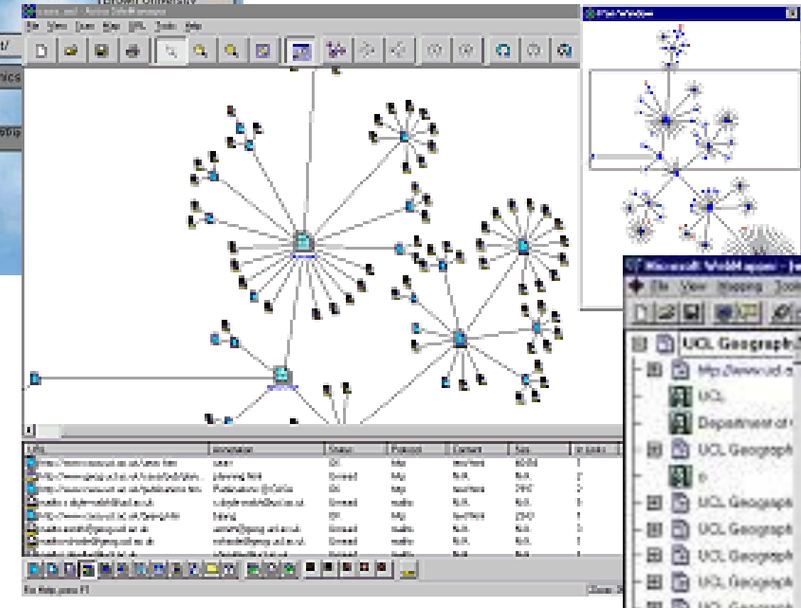

 copyright ©2025 g.di battista, m.patrignani

# Applications: websites analysis

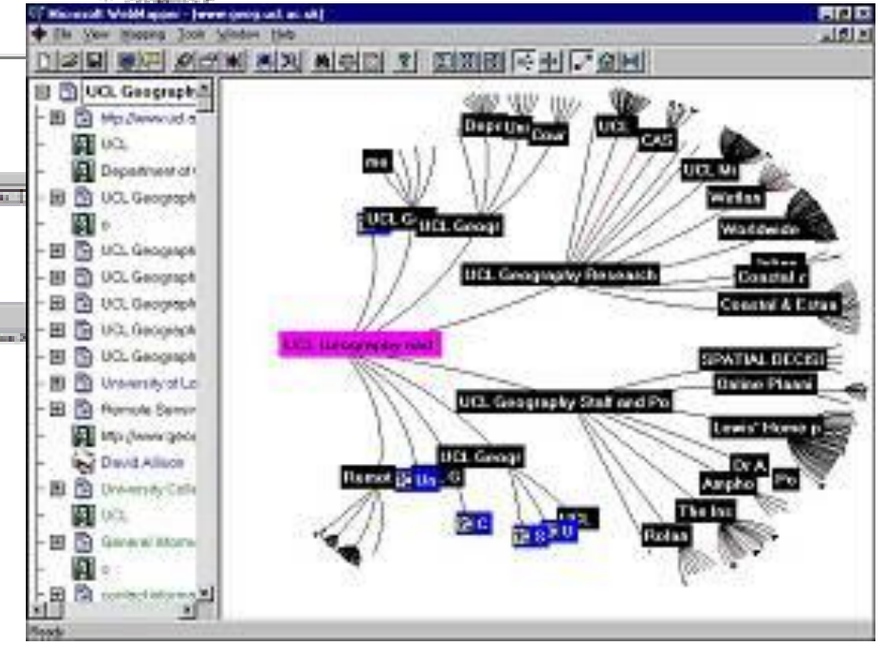
Ptolomaeus




Site Manager, Astra



Microsoft, Siteserver



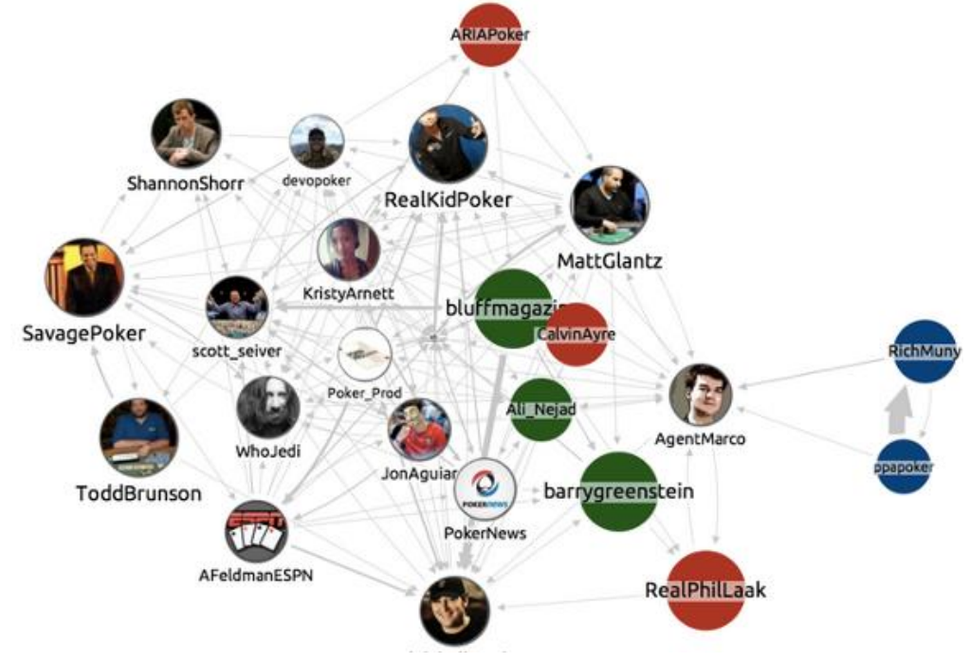

 copyright ©2025 g.di battista, m.patrignani

# Emerging applications

- Intelligence analysis (association charts)
- Forensic accounting (asset flow diagrams)
- Information security (attack graphs)
- Sociology (social network analysis)
- …



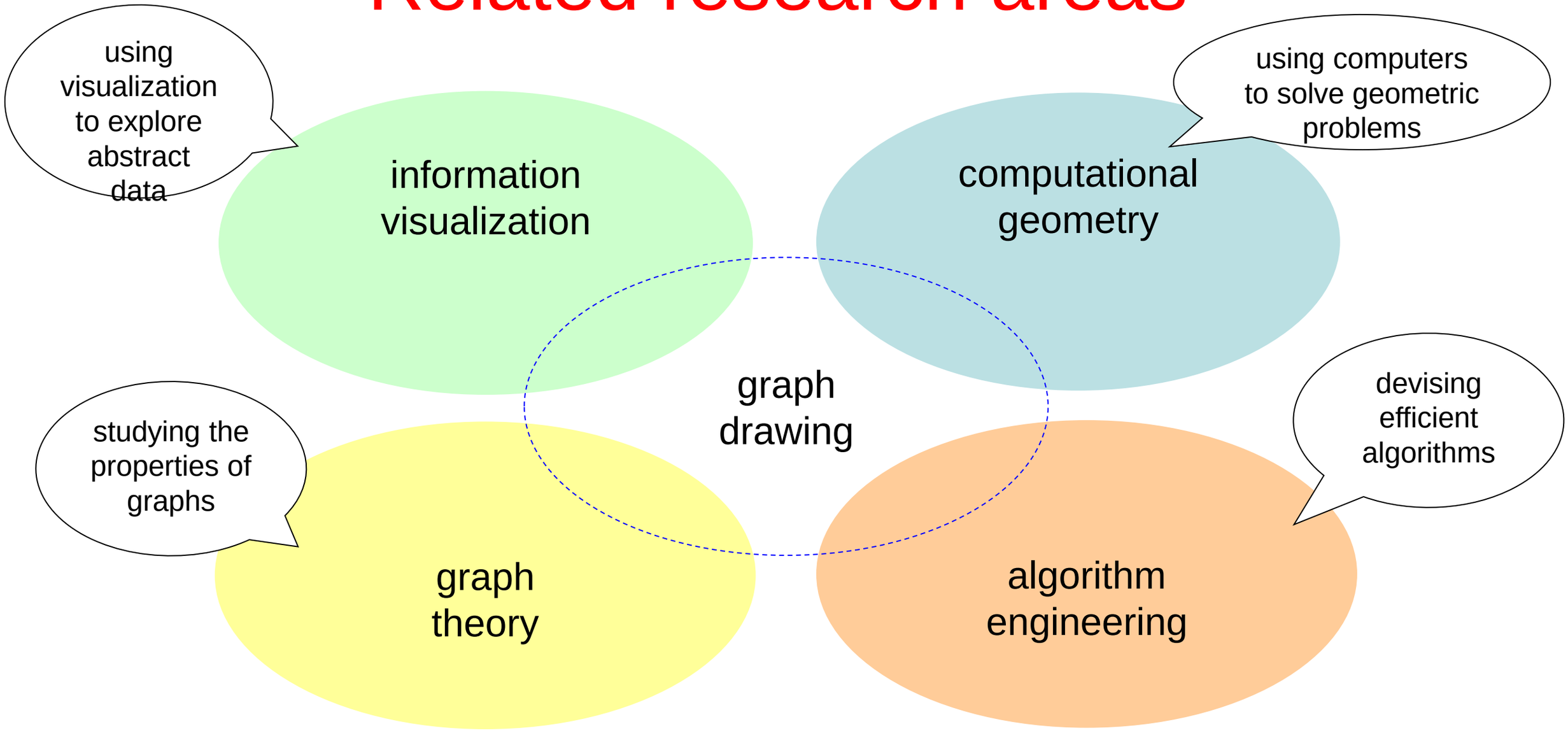

[zoomcharts.com]




[cambridge-intelligence.com]

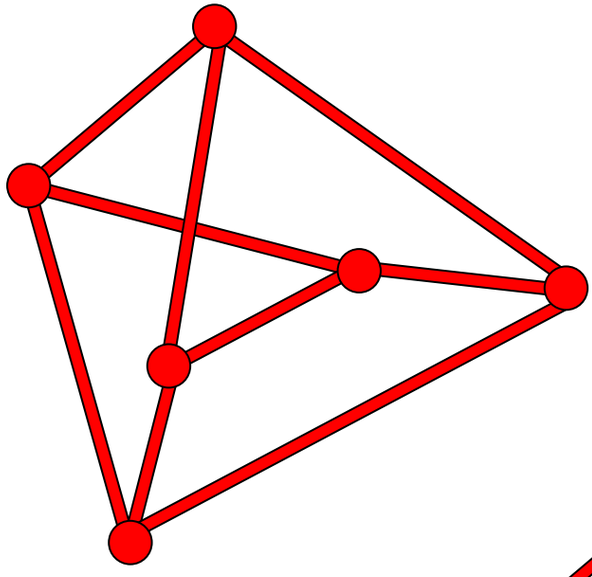
 copyright ©2025 g.di battista, m.patrignani

# Related research areas



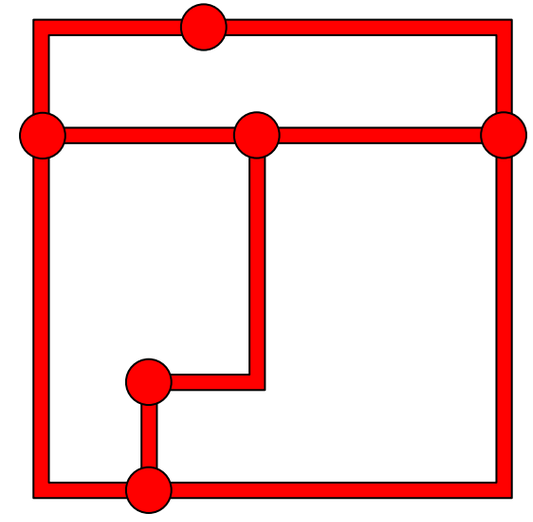

 copyright ©2025 g.di battista, m.patrignani

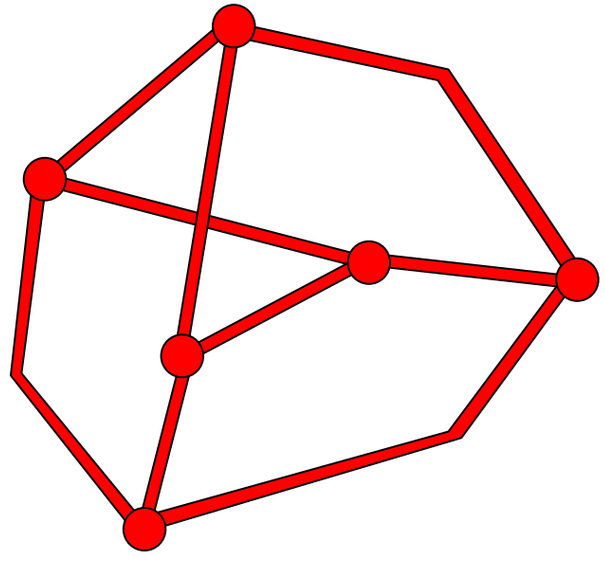
# Drawing conventions



straight-line

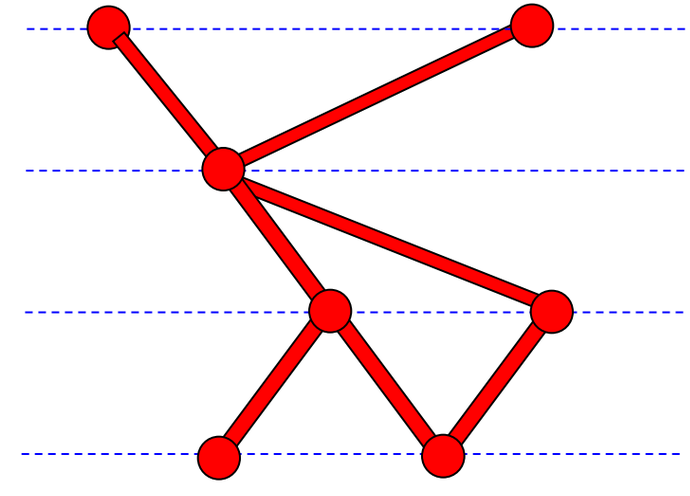
orthogonal





polyline

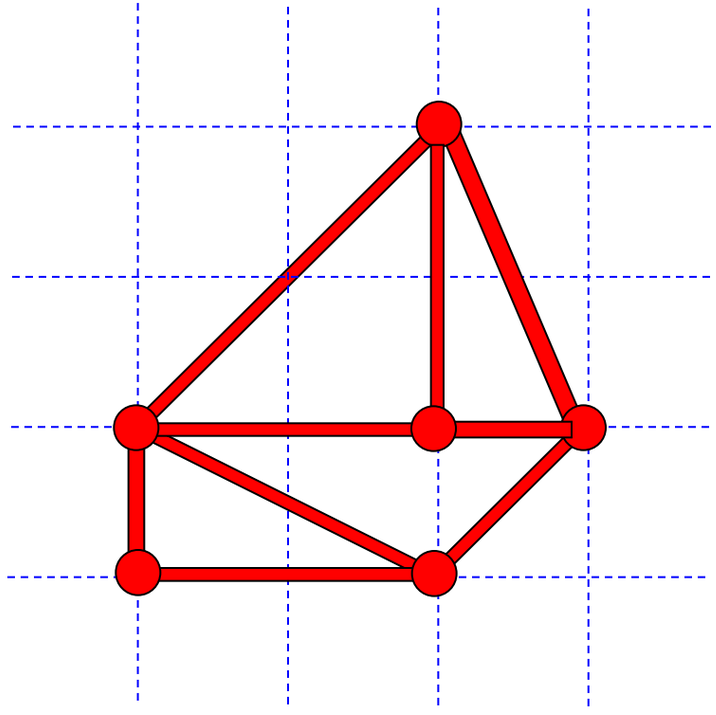
layered



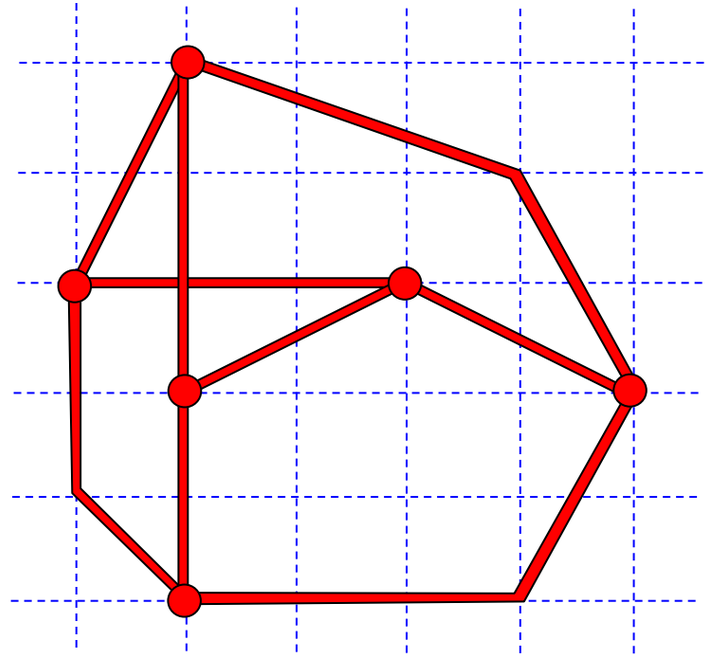

copyright ©2025 g.di battista, m.patrignani

# Drawings on the grid

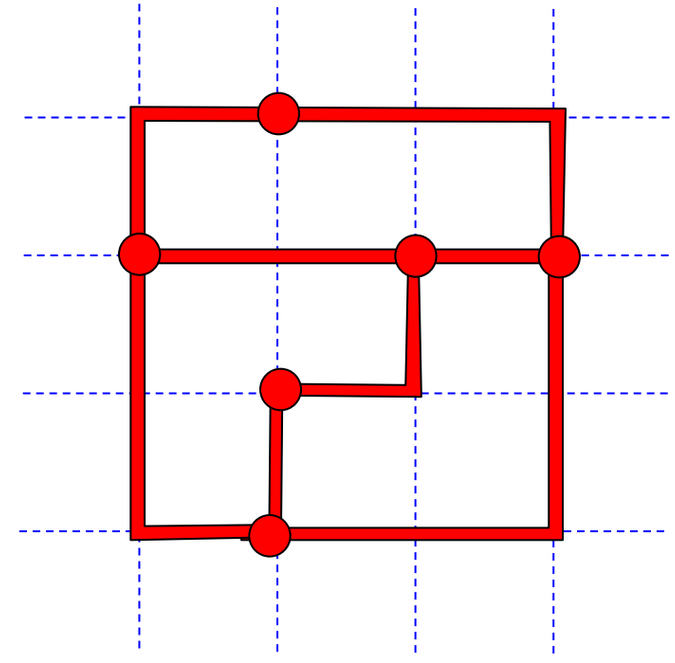
straight-line on the grid



polyline on the grid



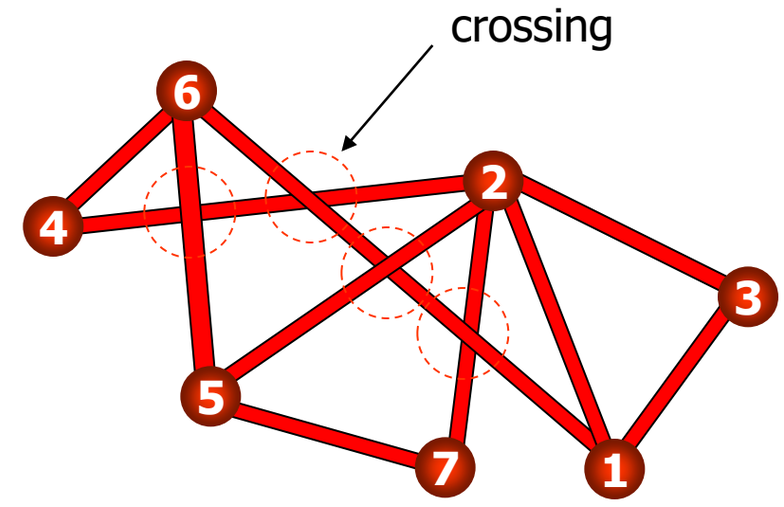
orthogonal on the grid

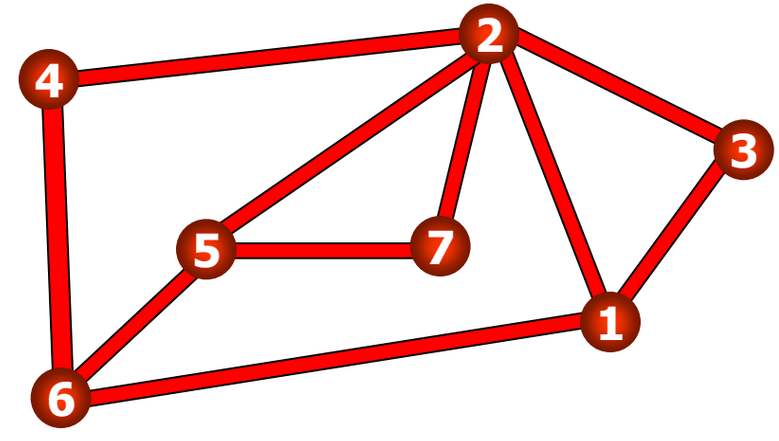



copyright ©2025 g.di battista, m.patrignani

# Aesthetics: few crossings

- The drawing is clearer if the number of intersections is reduced
  - intersections are discontinuities



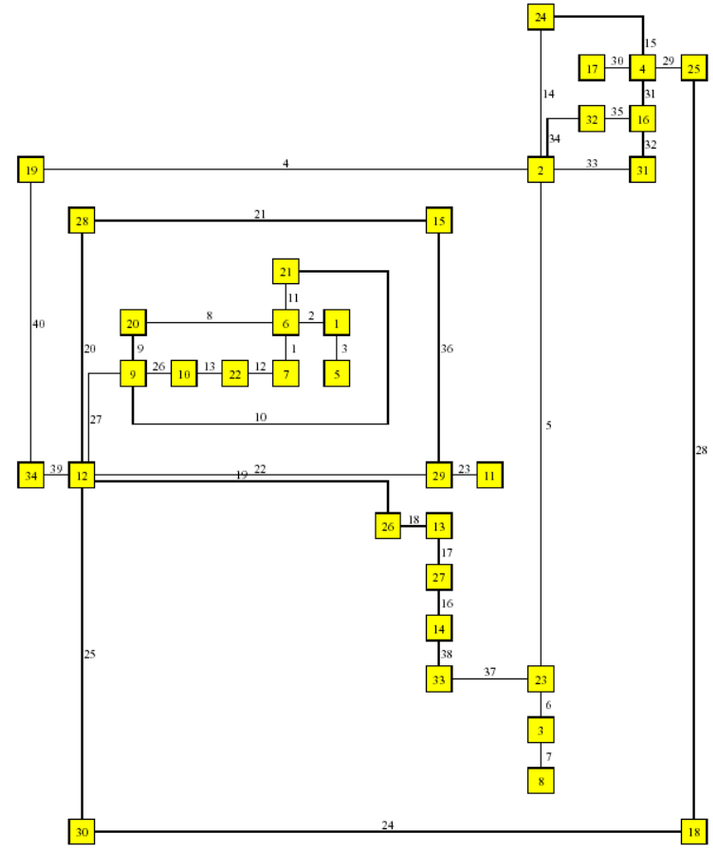



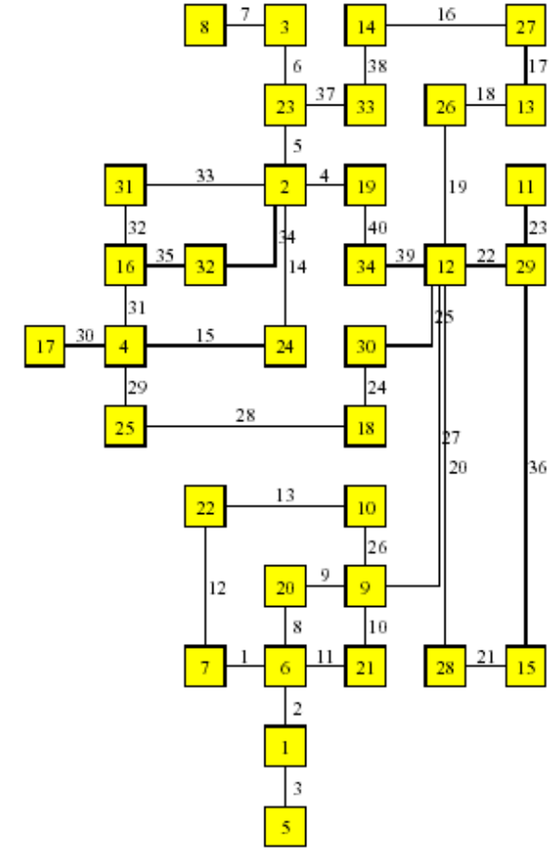

a non-planar and a planar drawing of the same graph

 copyright ©2025 g.di battista, m.patrignani

# Aesthetics: small size

- A small size allows to see a larger picture of the drawing
  - you may minimize the area, the maximum edge length, the average edge length, etc

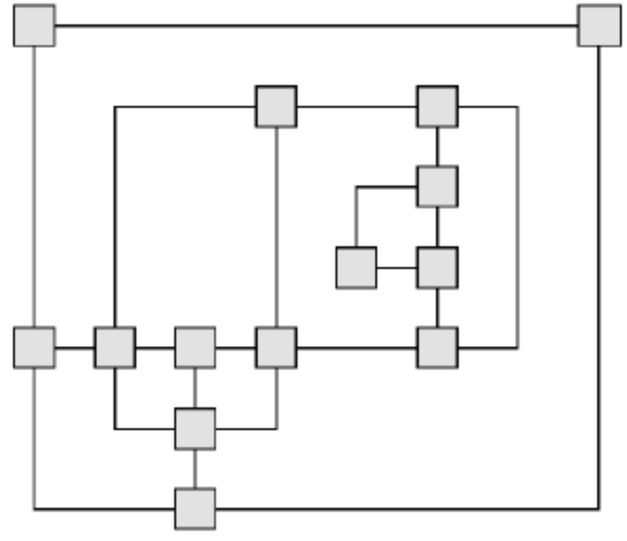




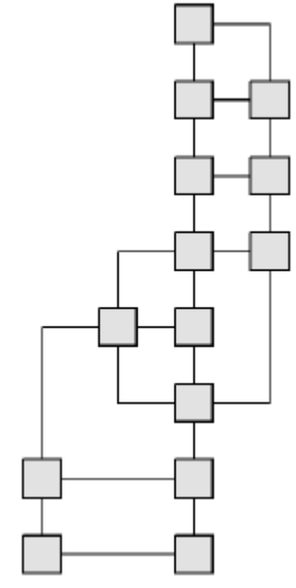
 copyright ©2025 g.di battista, m.patrignani

# Aesthetics: number of bends

- Edges with many bends are difficult to follow
  - based on the Gestalt law of continuity



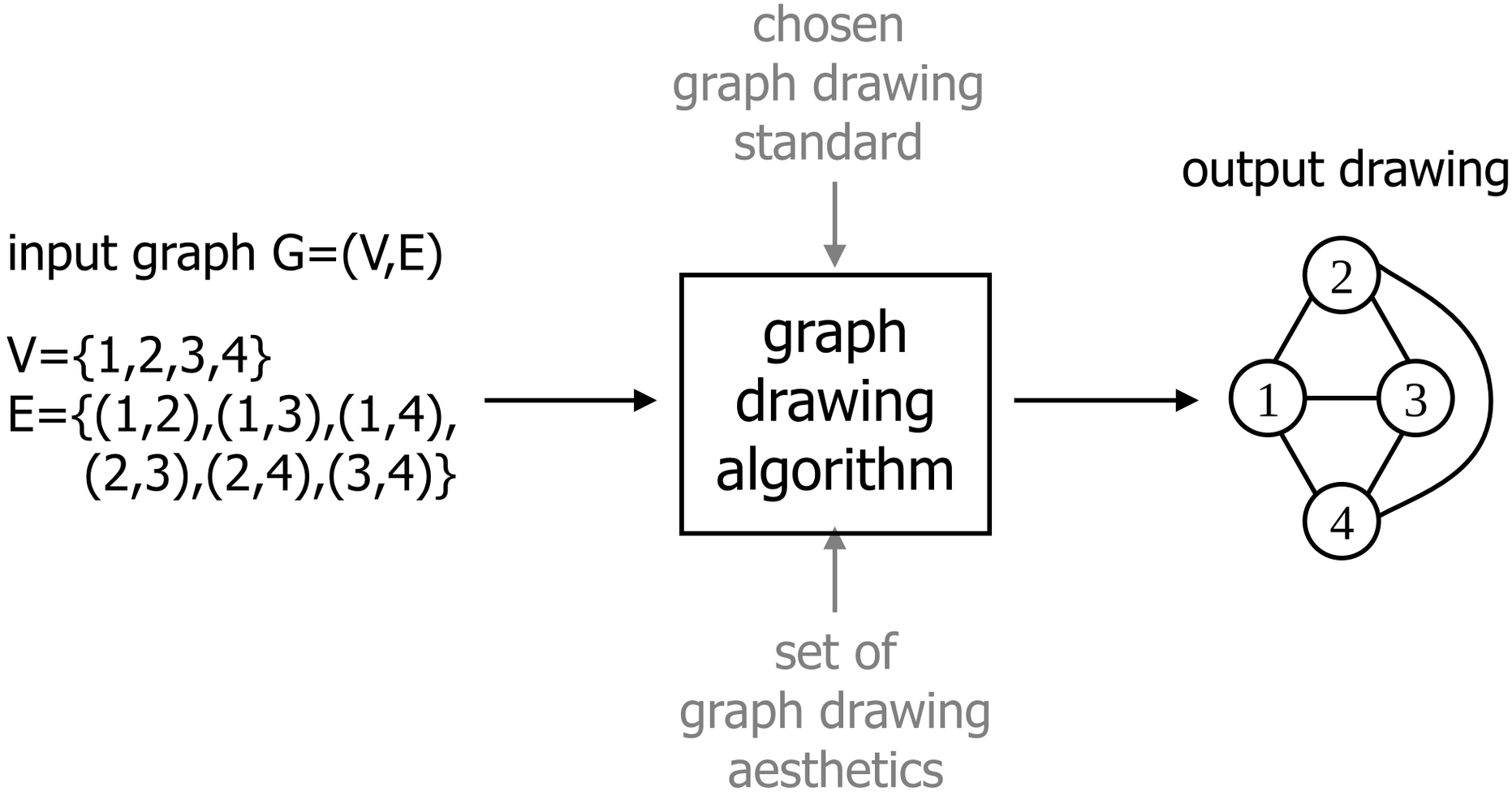
8 bends



5 bends

 copyright ©2025 g.di battista, m.patrignani

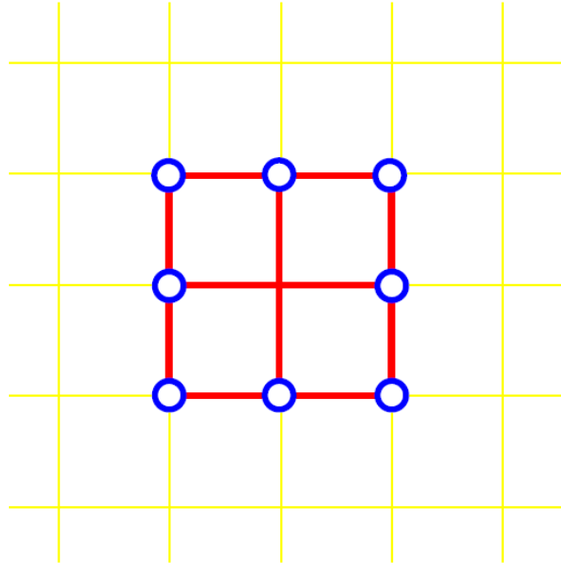
# Graph drawing purpose (revised)



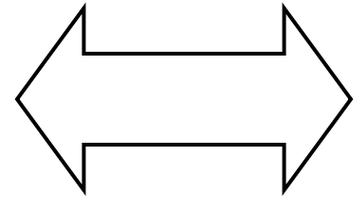

 copyright ©2025 g.di battista, m.patrignani

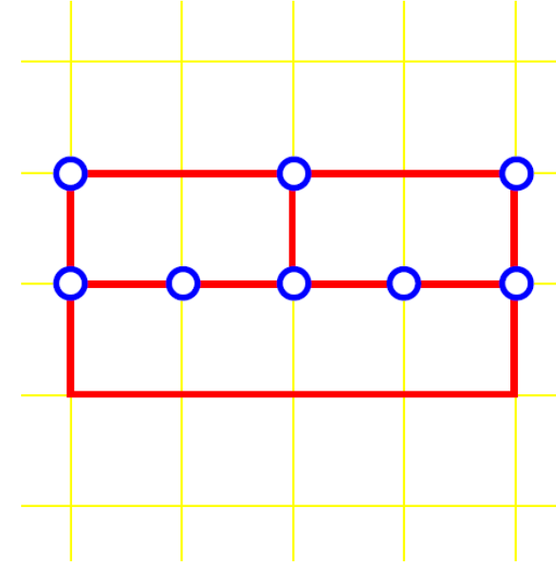
# Contrast of aesthetics

- Two drawings of the same graph within the “orthogonal on the grid” drawing convention



minimum number of bends





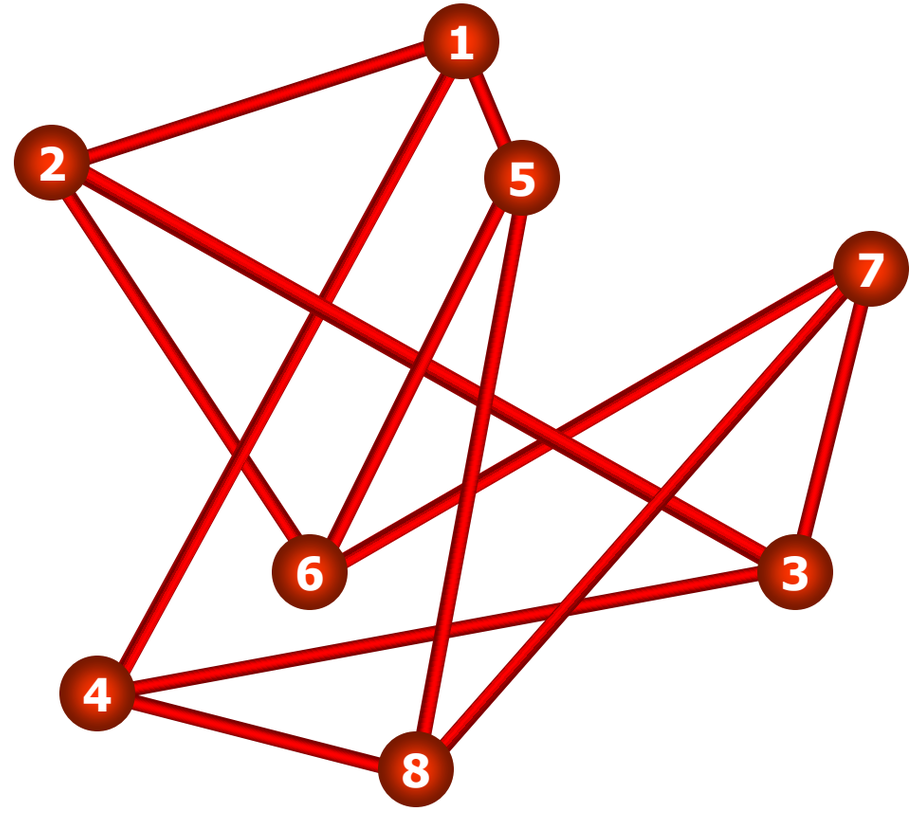
minimum number of crossings

Pictures from [Di Battista *et al.* 99]

 copyright ©2025 g.di battista, m.patrignani

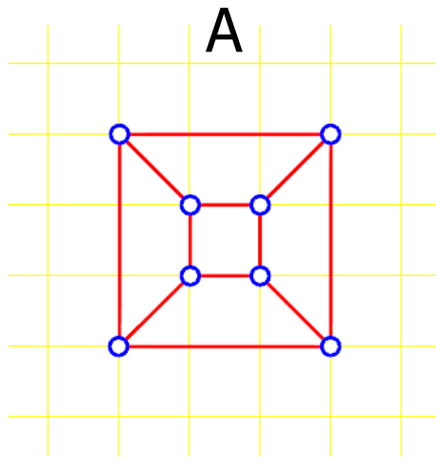
# Graph drawing exercise

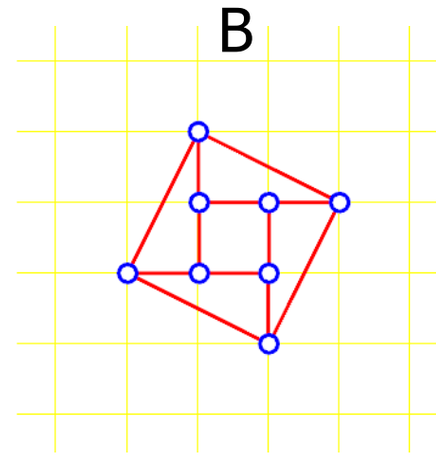
- Produce a nice straight-line drawing (on the grid) of this graph

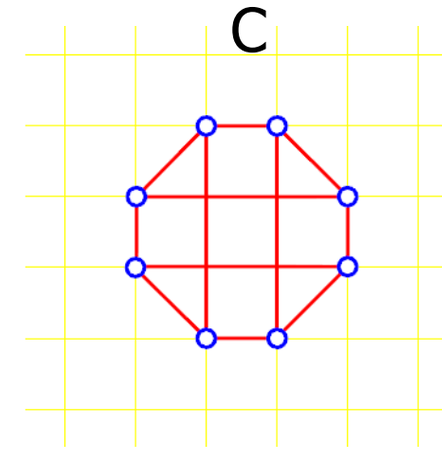


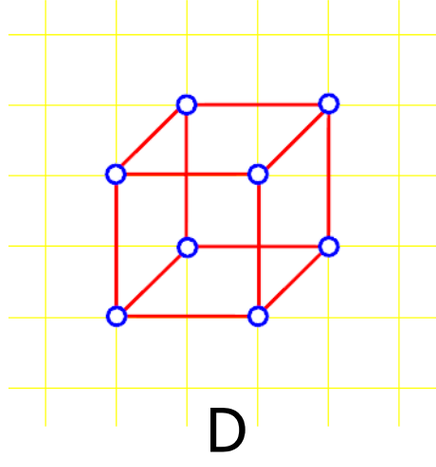

 copyright ©2025 g.di battista, m.patrignani

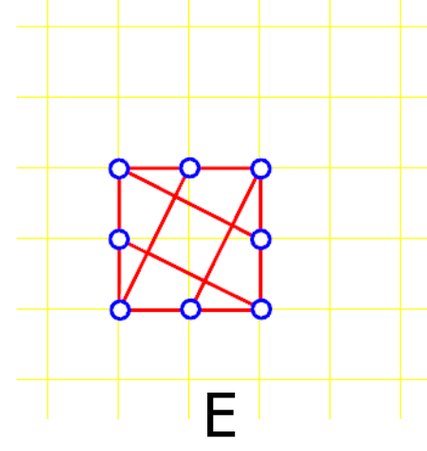
# Some possible answers

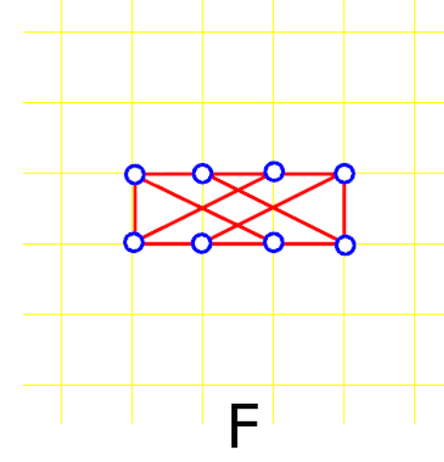


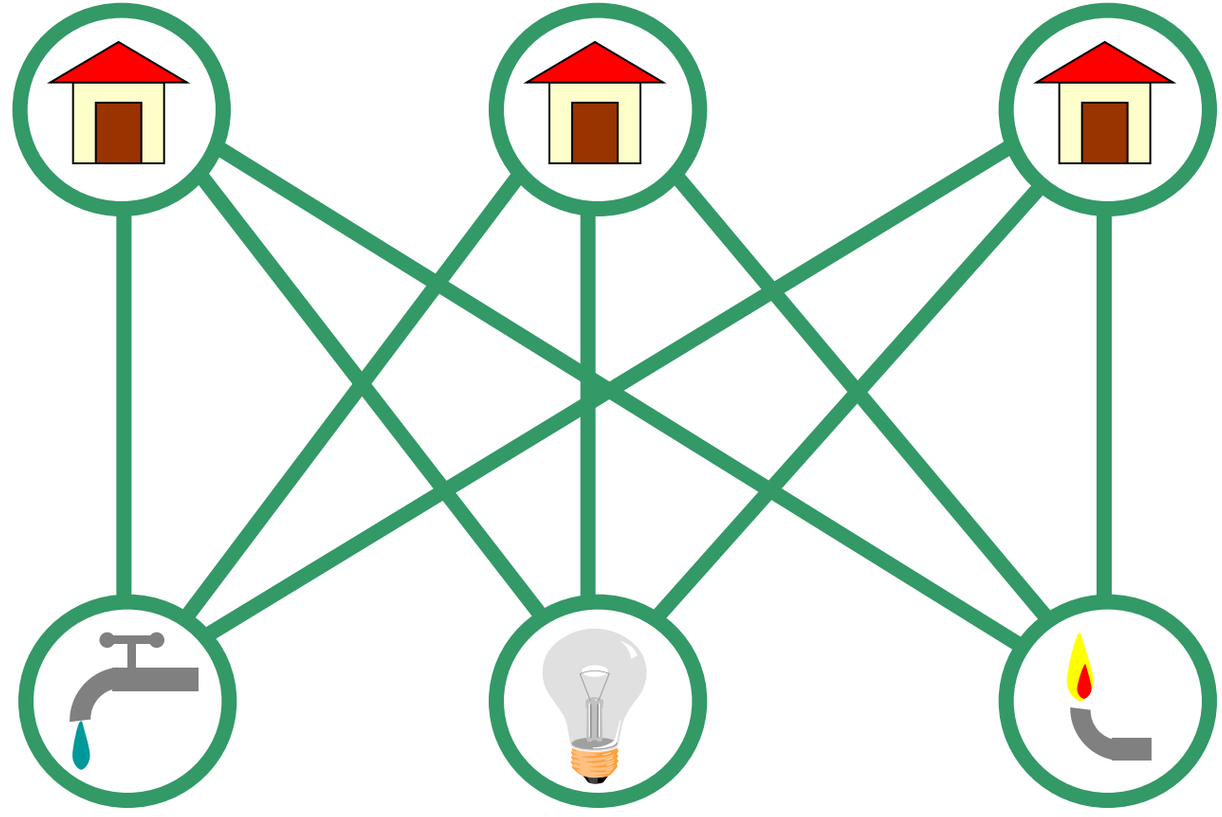




Pictures from [Di Battista *et al.* 99]

 copyright ©2025 g.di battista, m.patrignani

# Graph drawing exercise

- Draw this graph without intersections



solution: it is not possible

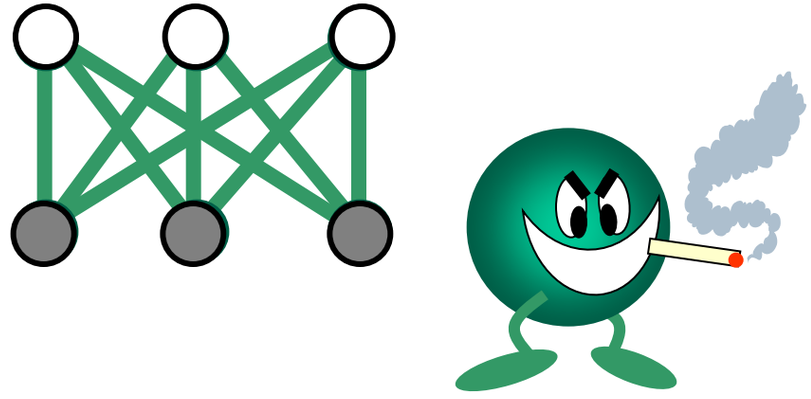

copyright ©2025 g.di battista, m.patrignani

# Non-planar graphs

- A drawing is planar if it does not have intersections (edge crossings)
- A graph is planar if it admits a planar drawing
  - not all graphs are planar

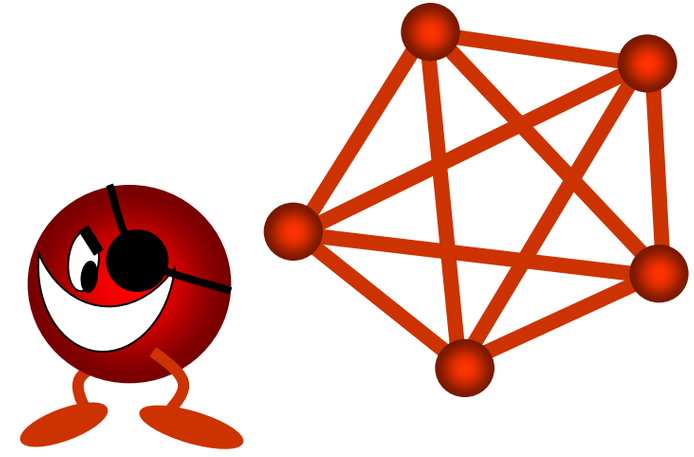
 copyright ©2025 g.di battista, m.patrignani

# Two (in)famous non-planar graphs



$K_{3,3}$

complete bipartite
graph with six vertices



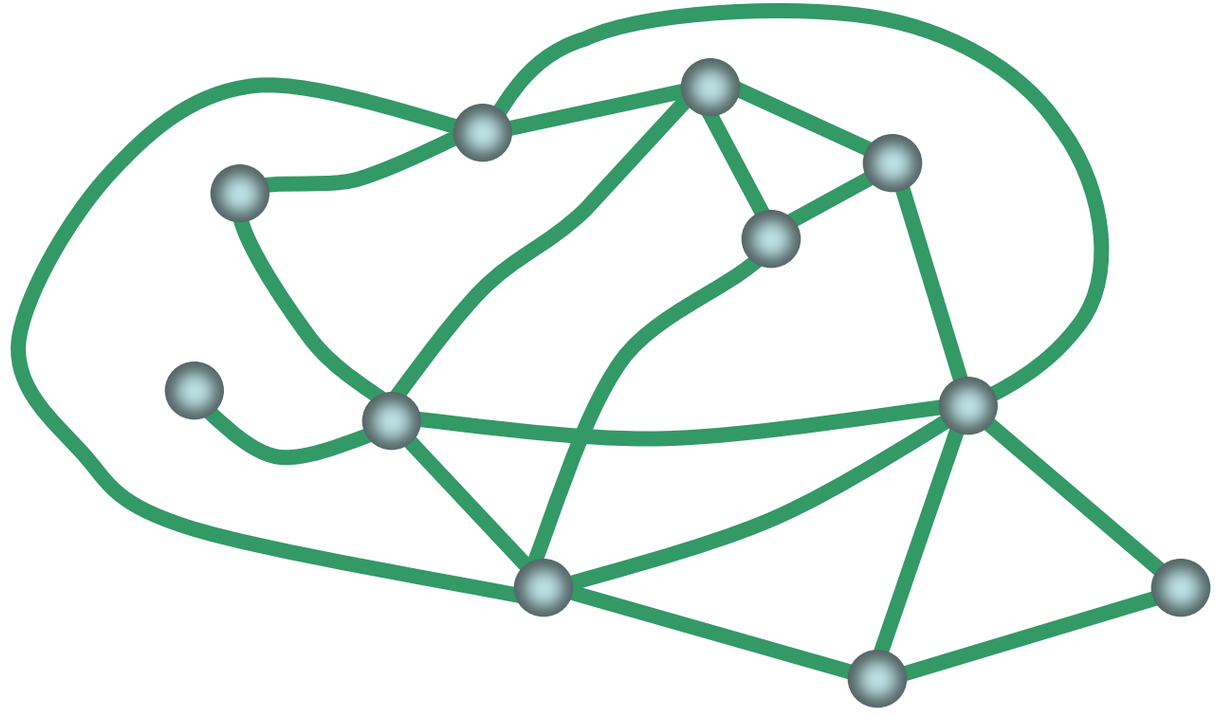
$K_5$

complete graph with
five vertices


copyright ©2025 g.di battista, m.patrignani

# Characterization of non-planar graphs

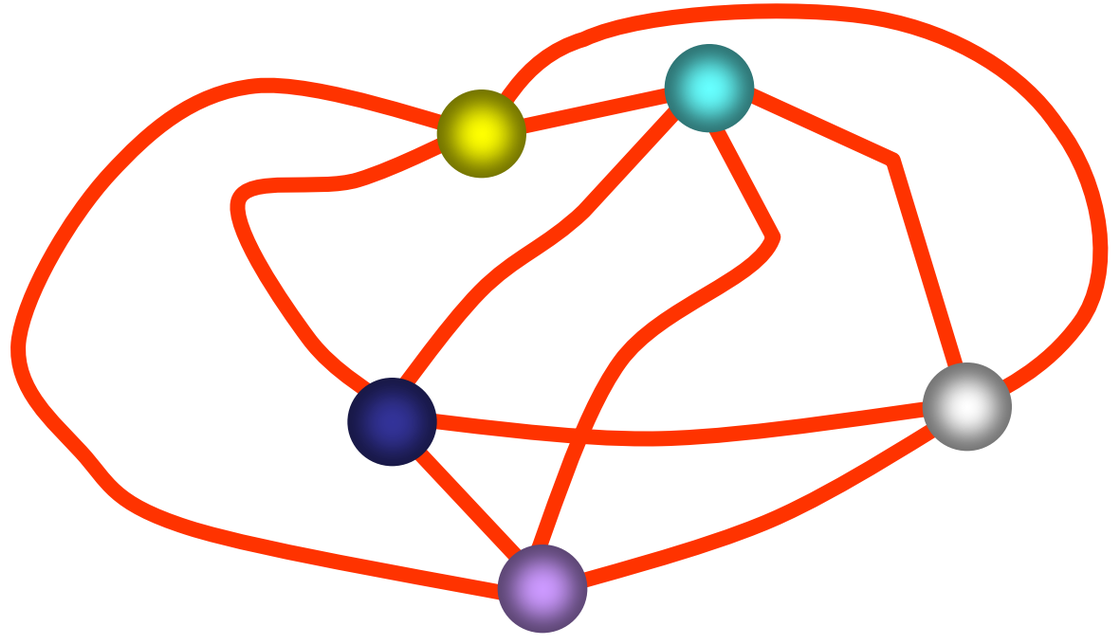
- [Kuratowsky 30] states
  - a graph is non-planar if and only if it contains a subdivision of $K_5$ or $K_{3,3}$
- What is a “subdivision”?

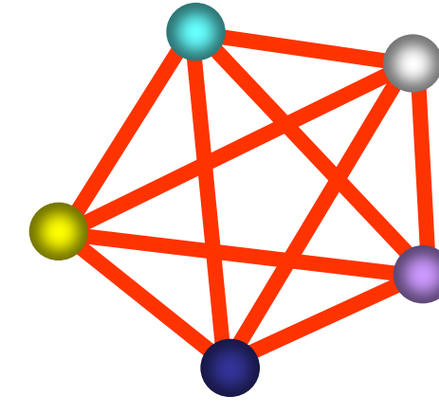


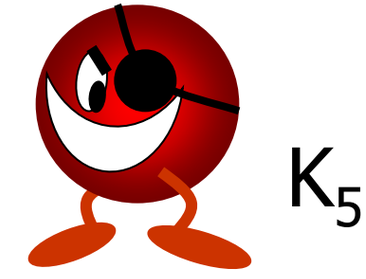
 copyright ©2025 g.di battista, m.patrignani

# Characterization of non-planar graphs

- [Kuratowsky 30] states
  - a graph is non-planar if and only if it contains a subdivision of $K_5$ or $K_{3,3}$
- What is a “subdivision”?



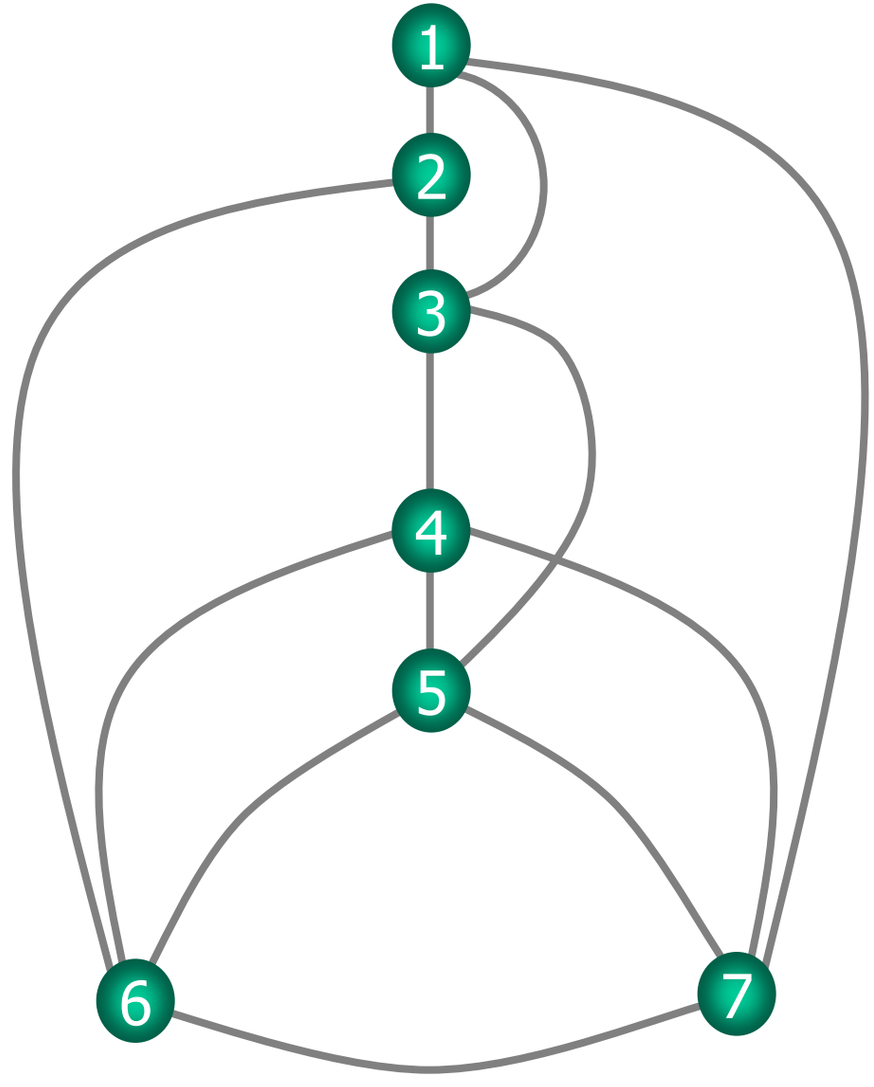




$K_5$

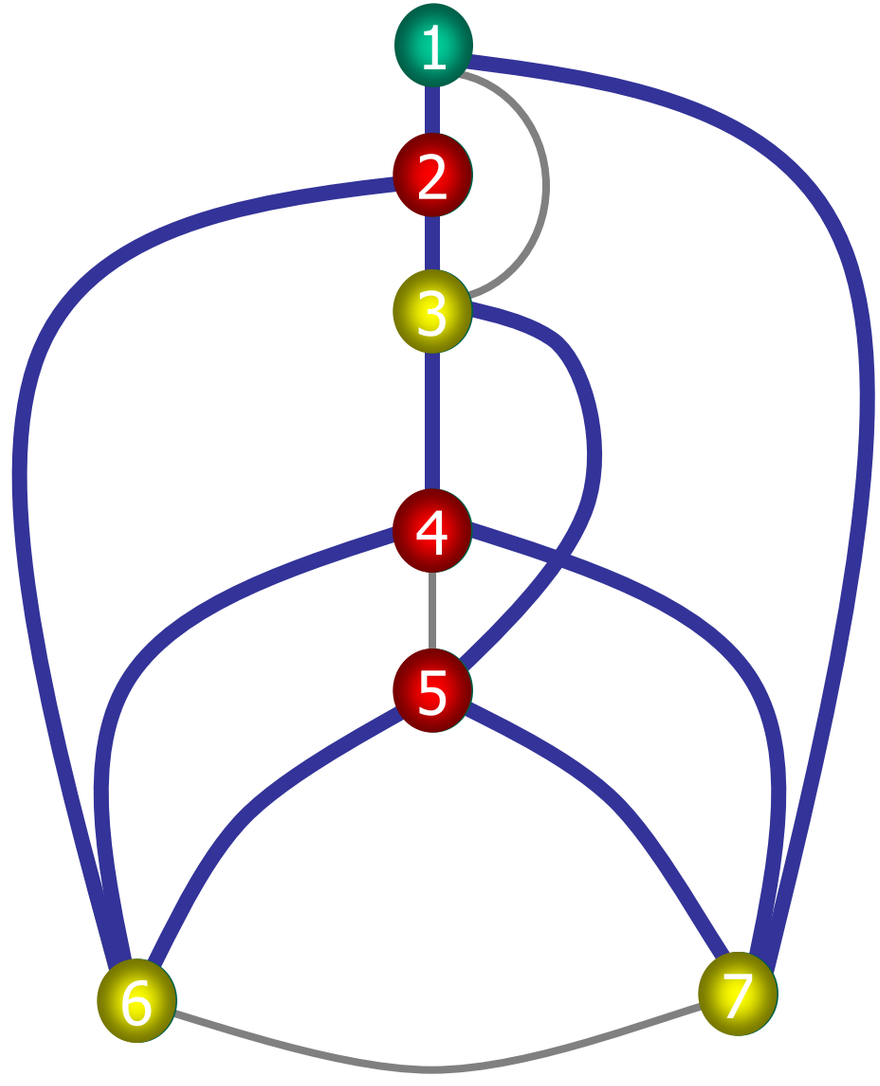

copyright ©2025 g.di battista, m.patrignani

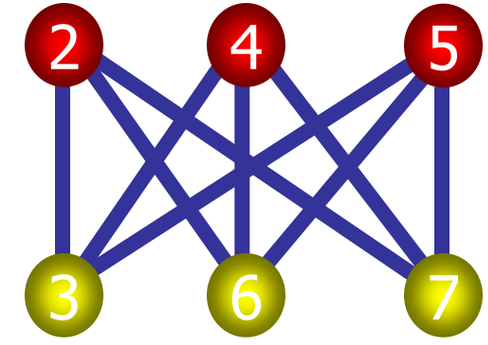
# Exercise: find the $K_{3,3}$ subdivision




 copyright ©2025 g.di battista, m.patrignani

# Exercise: find the $K_{3,3}$ subdivision






$K_{3,3}$

 copyright ©2025 g.di battista, m.patrignani

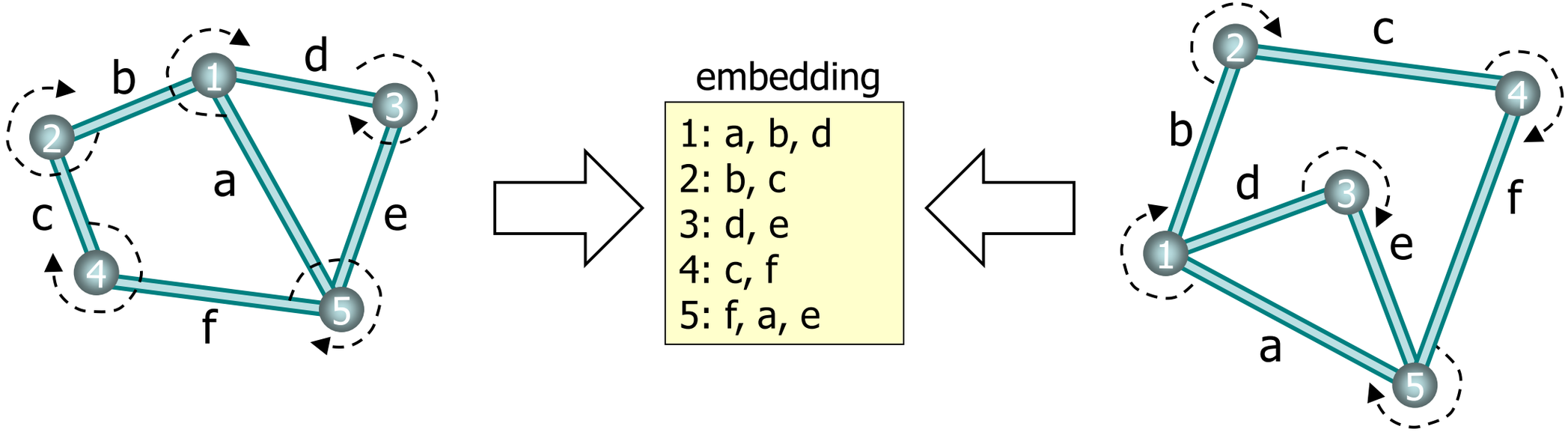
# Kuratowsky subgraphs

- A *subdivision of a graph* (for example a $K_{3,3}$ or a $K_5$ graph) is obtained from the graph itself by replacing some edges with some paths
- Unfortunately, searching for subdivisions of $K_{3,3}$ or $K_5$ does not lead to an efficient method to test the planarity of a graph
  - to search for subdivisions of $K_5$, for example, you should consider all sets of 5 nodes and verify the existence of independent paths among them

 copyright ©2025 g.di battista, m.patrignani

# The concept of embedding

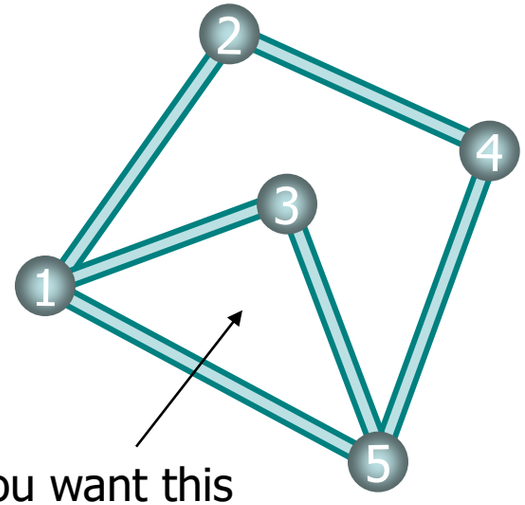
- The circular order of the edges around each vertex is called *embedding*
  - these two drawings have the same embedding



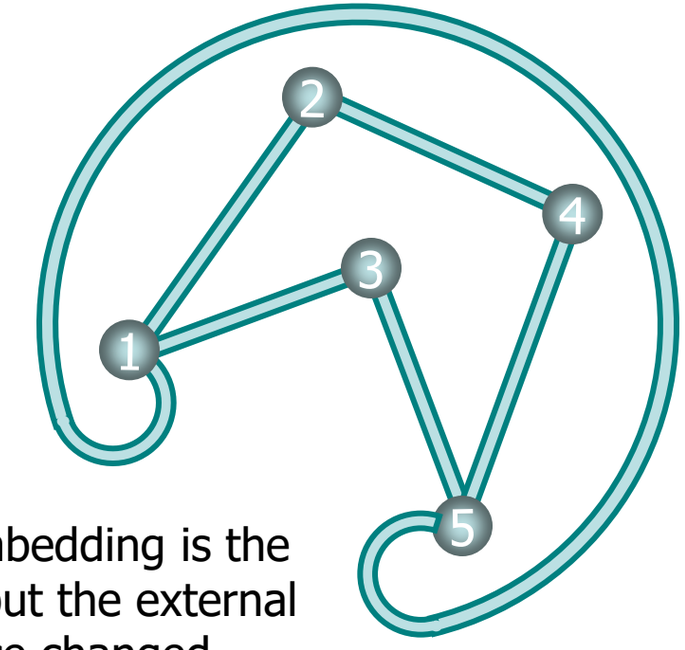

 copyright ©2025 g.di battista, m.patrignani

# Embeddings and external face

- Any face of the graph can be made the external one within the same planar embedding




you want this face to be the external one?




the embedding is the same but the external face changed

 copyright ©2025 g.di battista, m.patrignani

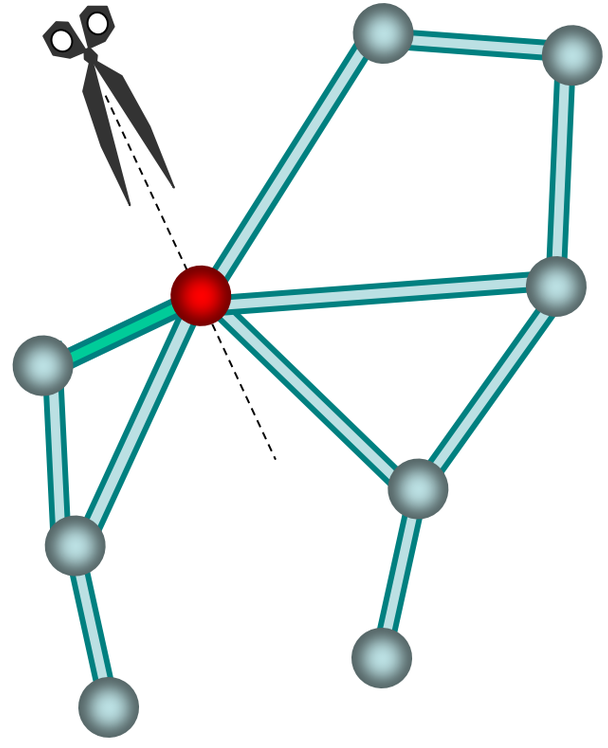
# Finding a planar embedding

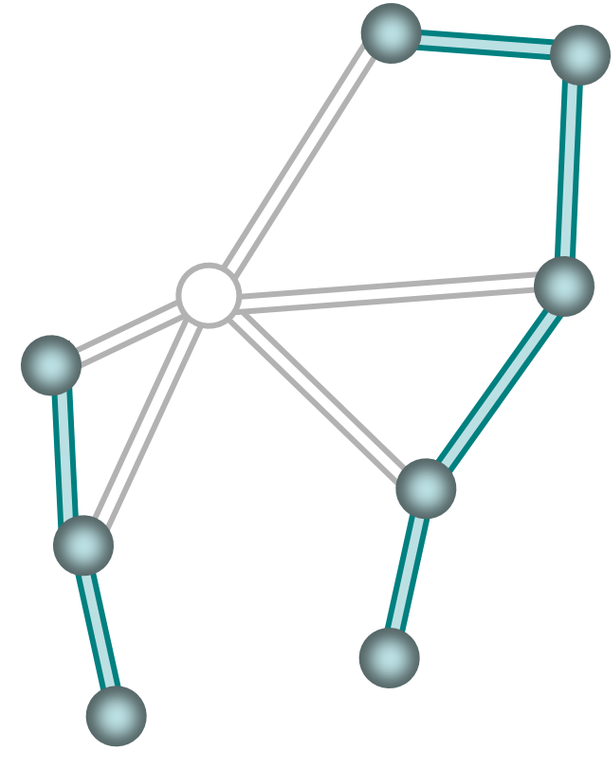
- We focus on the problem of finding a planar embedding
  - once a planar embedding is known there are several algorithms to compute a planar drawing of the graph
- We will use a divide-and-conquer strategy

 copyright ©2025 g.di battista, m.patrignani

# Cut vertices

- A *cutvertex* is a vertex such that its removal produces a disconnected graph

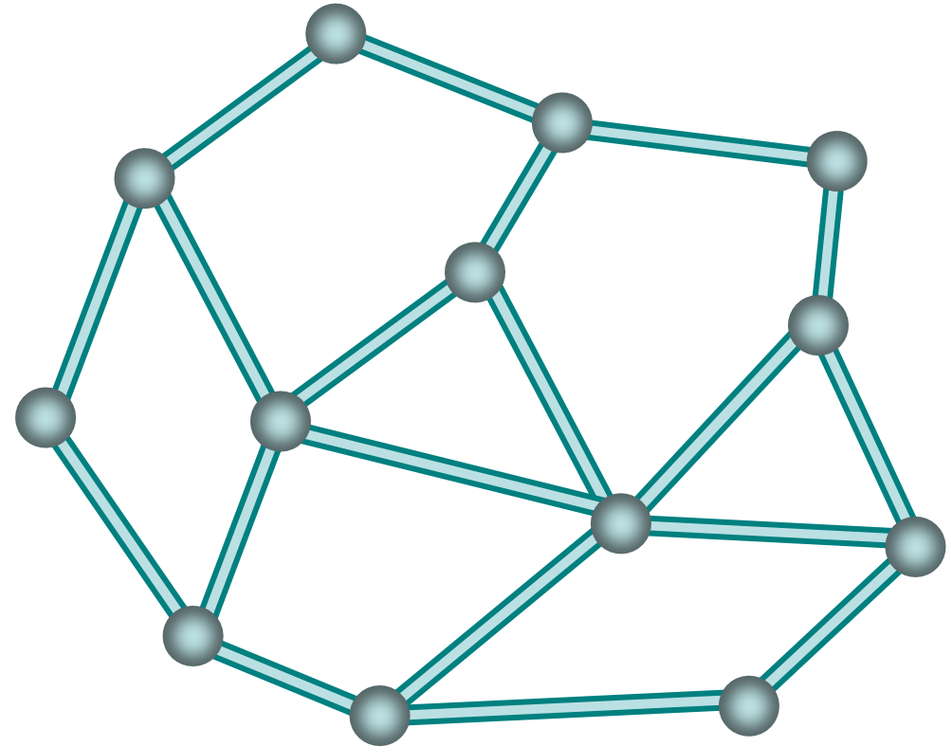




 copyright ©2025 g.di battista, m.patrignani

# Exercise

- Find all the cutvertices of this graph

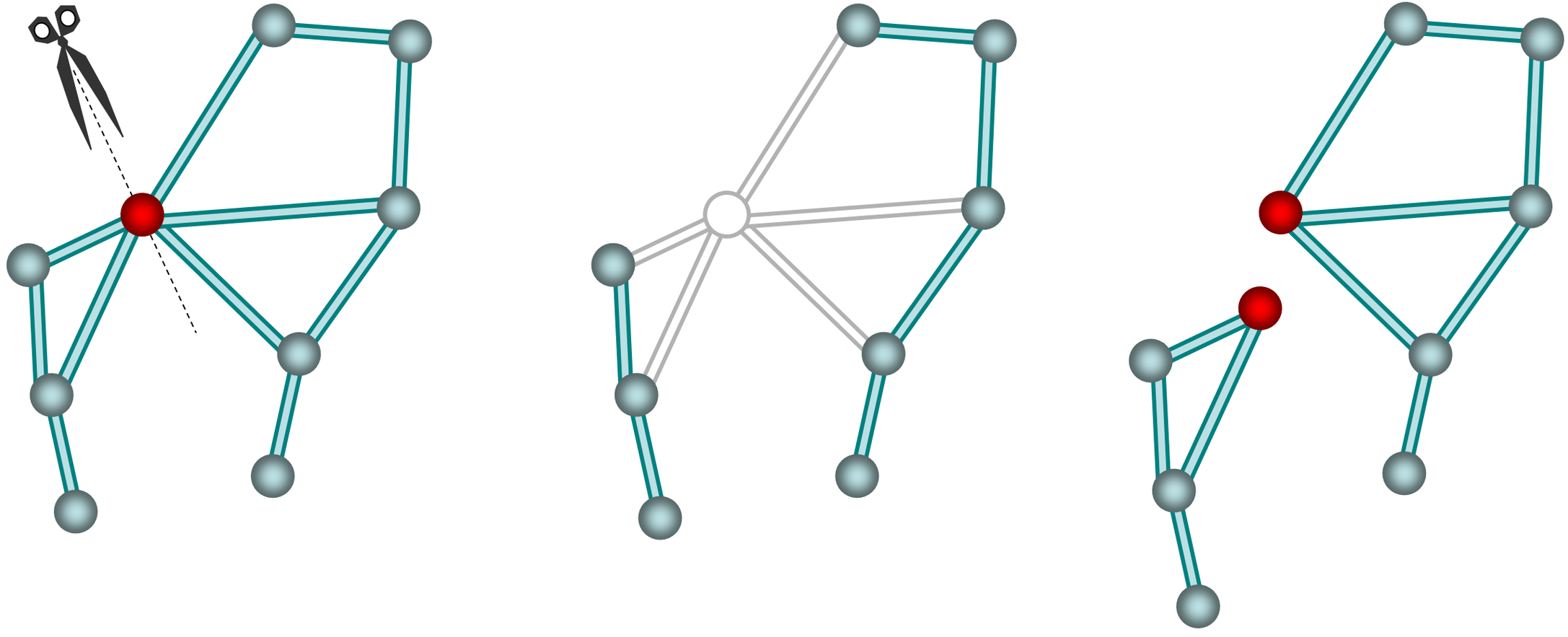


when there is no cutvertex the graph is said to be *biconnected*

 copyright ©2025 g.di battista, m.patrignani

# Splitting the graph

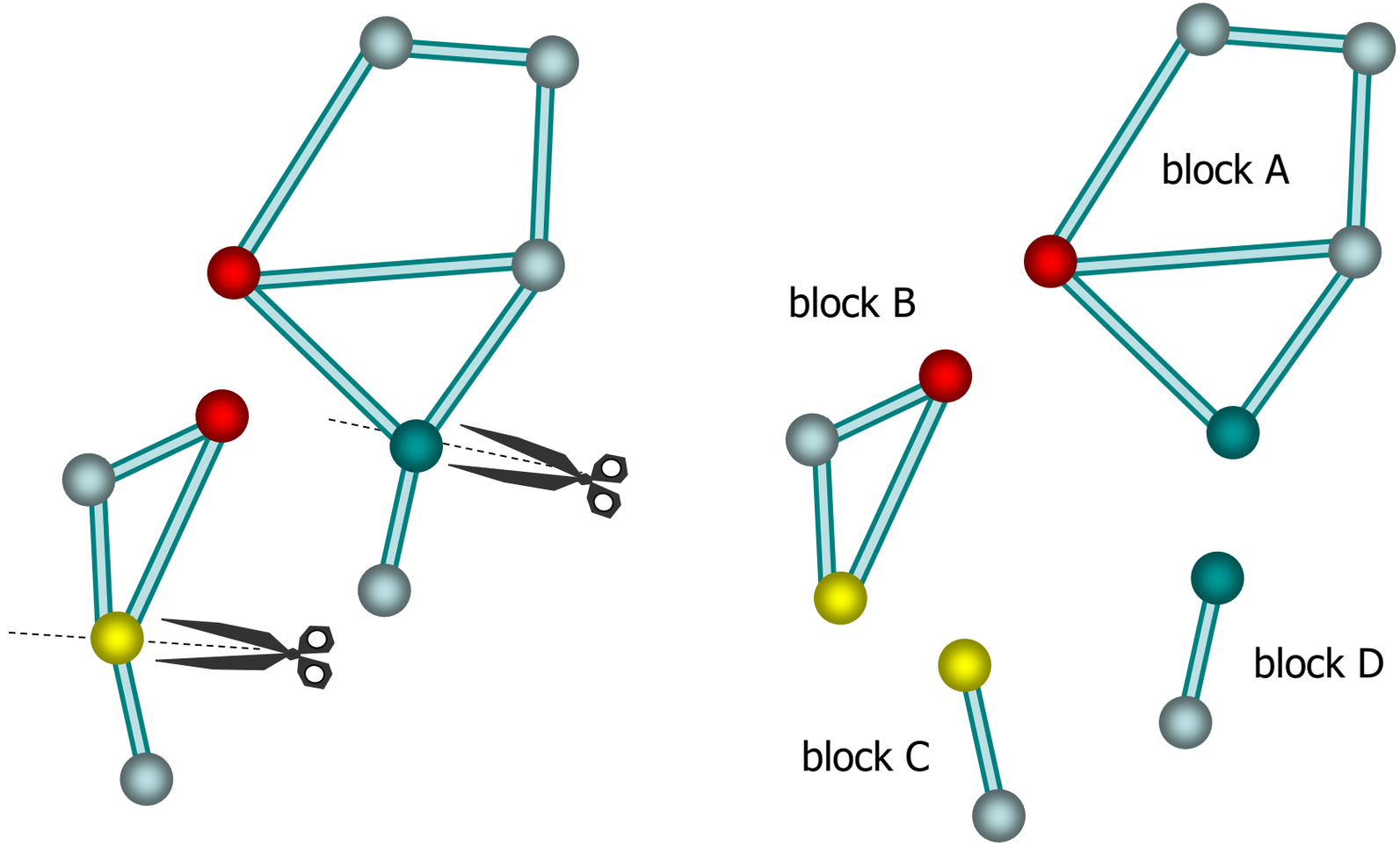
- Cutvertices can be used to split the graph
  - it is usual to leave a copy of the cutvertex in each component



 copyright ©2025 g.di battista, m.patrignani

# Splitting the graph

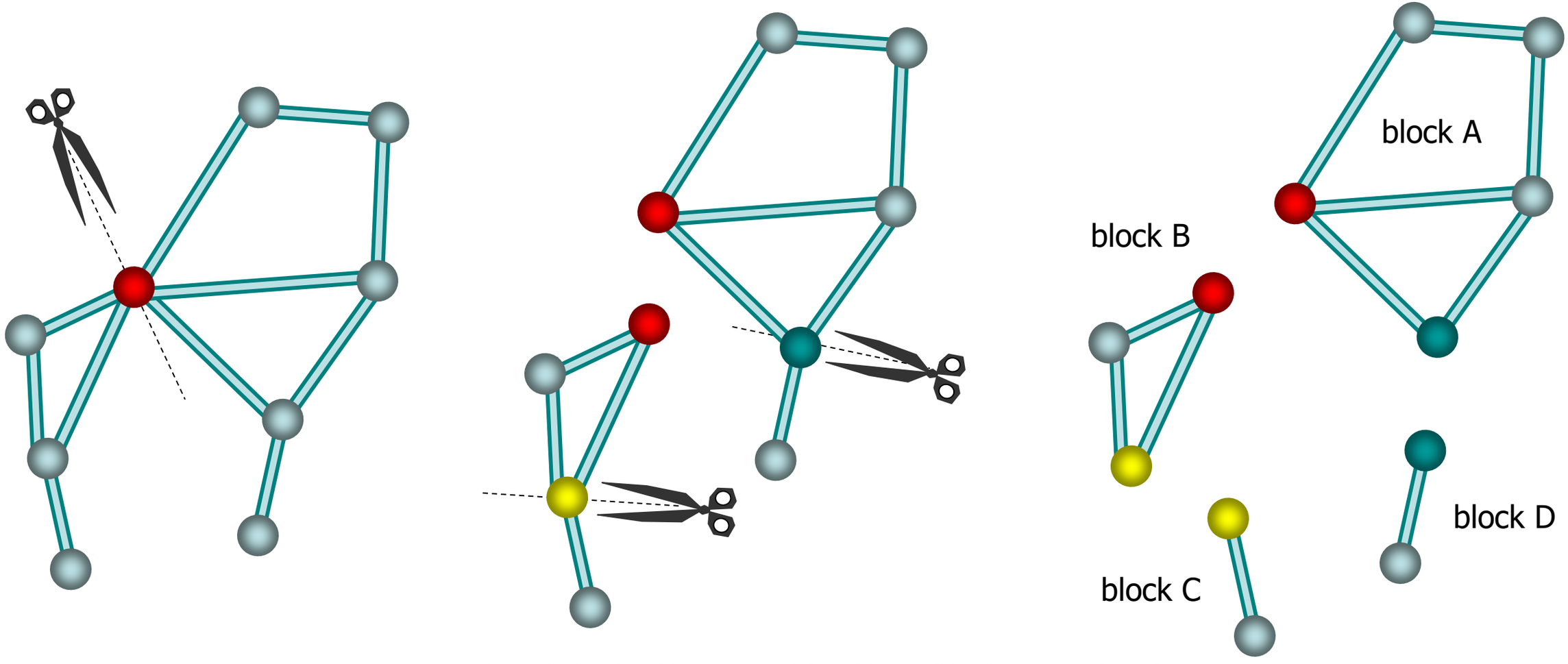
- You can carry on splitting the graph until it is possible




 copyright ©2025 g.di battista, m.patrignani

# Splitting the graph

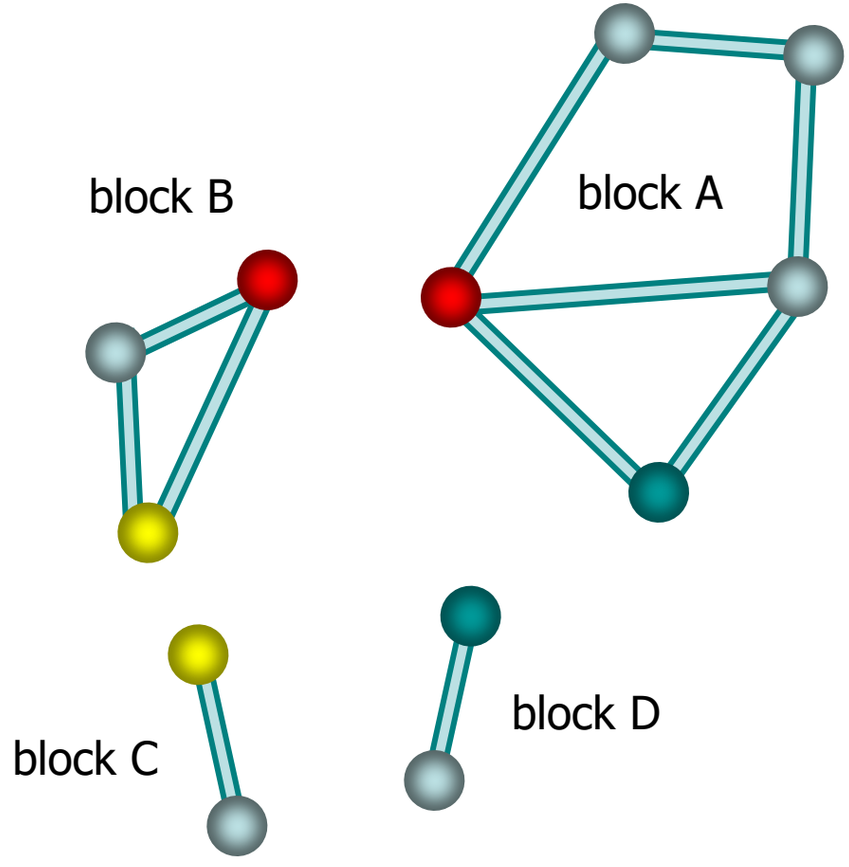
- recursively splitting a graph produces a set of biconnected pieces called *blocks*

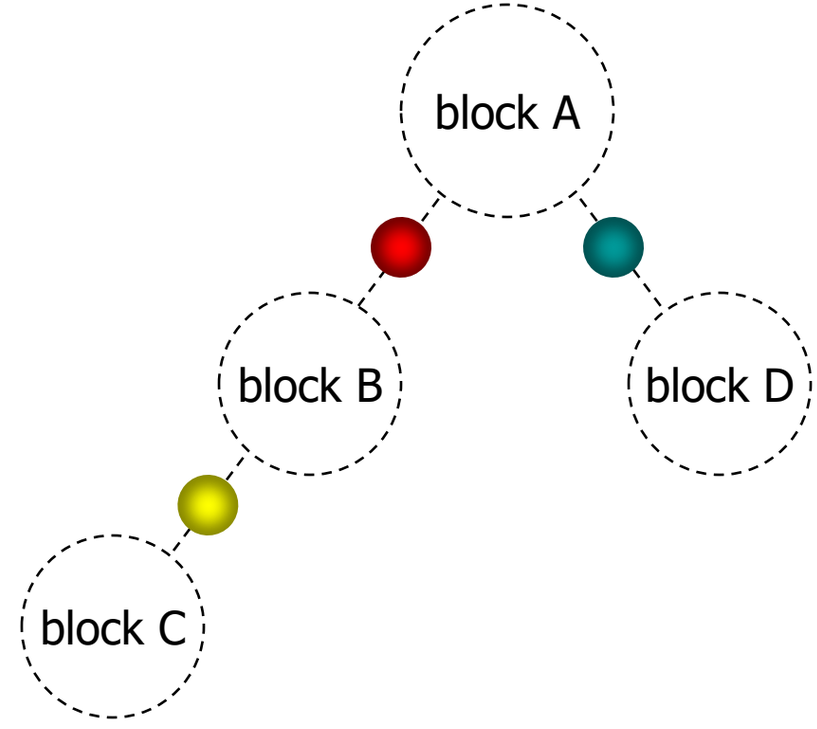



 copyright ©2025 g.di battista, m.patrignani

# Block-cutvertex tree

- The relationships among blocks and cutvertices can be described by a *block-cutvertex tree* (BC-tree)




 copyright ©2025 g.di battista, m.patrignani

# Decomposition exercise

- Count cutvertices and blocks



solution: there are four cutvertices and five blocks

 copyright ©2025 g.di battista, m.patrignani

# Decomposition exercise

- Count cut-vertices and blocks



solution: there are six cutvertices and eleven blocks

 copyright ©2025 g.di battista, m.patrignani

# Planarity of blocks and graphs

- A graph is planar if and only if its blocks are planar
- Proof:
  - if the graph is planar
    - take a planar drawing of it
    - all the blocks are planarly drawn
  - if all blocks are planar
    - consider the block-cut-vertex tree
    - planarly embed each block with the parent vertex on the external face
    - attach the blocks together to obtain a planar embedding of the whole graph

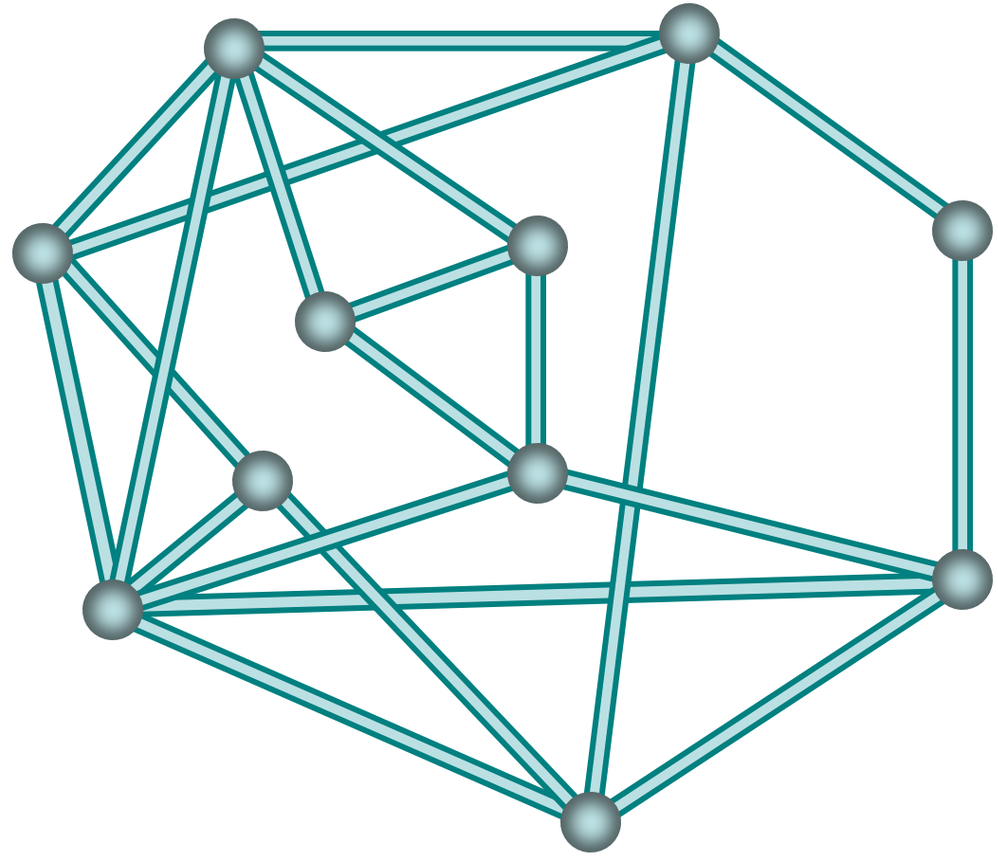
 copyright ©2025 g.di battista, m.patrignani

# A very simple algorithm

- Auslander and Parter algorithm [Auslander & Parter 61]
  - very old but very clear
  - simple but not linear (cubic)
  - we will only see the part that decides whether the input graph is planar
    - actually computing a planar embedding is more complicated
- Refined by [Hopcroft & Tarjan 74]
  - first linear-time algorithm for planarity
  - very complex
  - uses DFS
  - the embedding phase of this algorithm was described only in 1996 by [Mehlhorn & Mutzel 96]

 copyright ©2025 g.di battista, m.patrignani

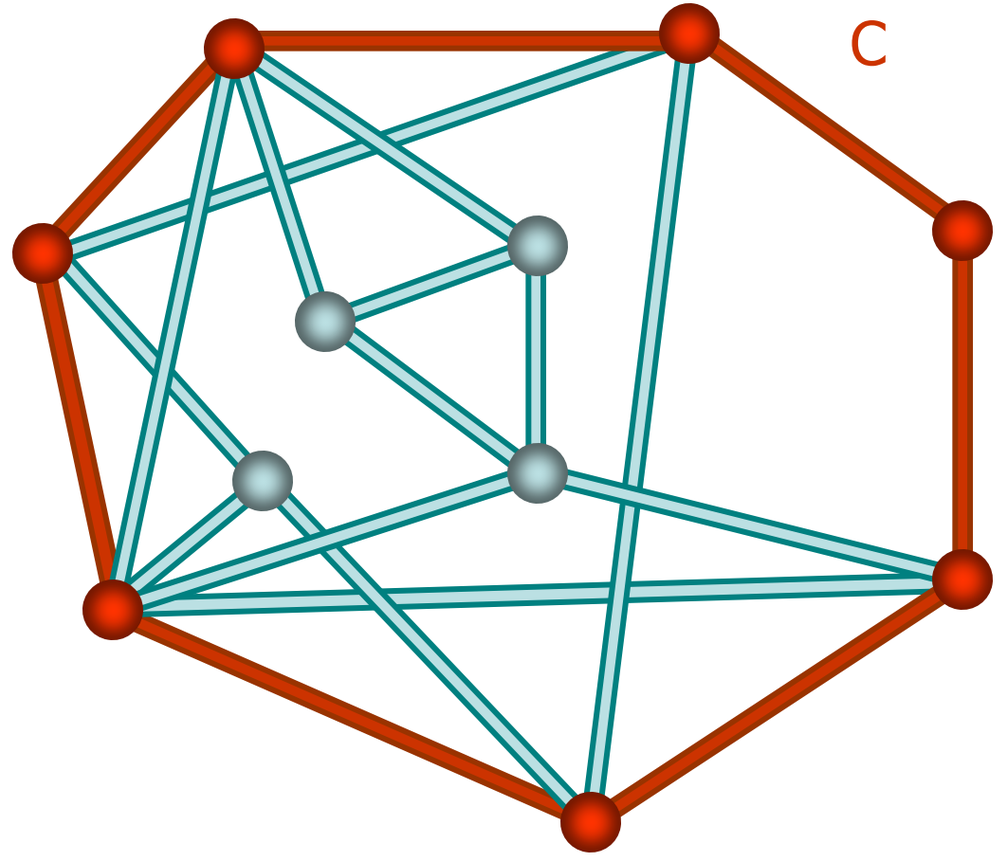
# Input graph (assumed to be biconnected)



 copyright ©2025 g.di battista, m.patrignani

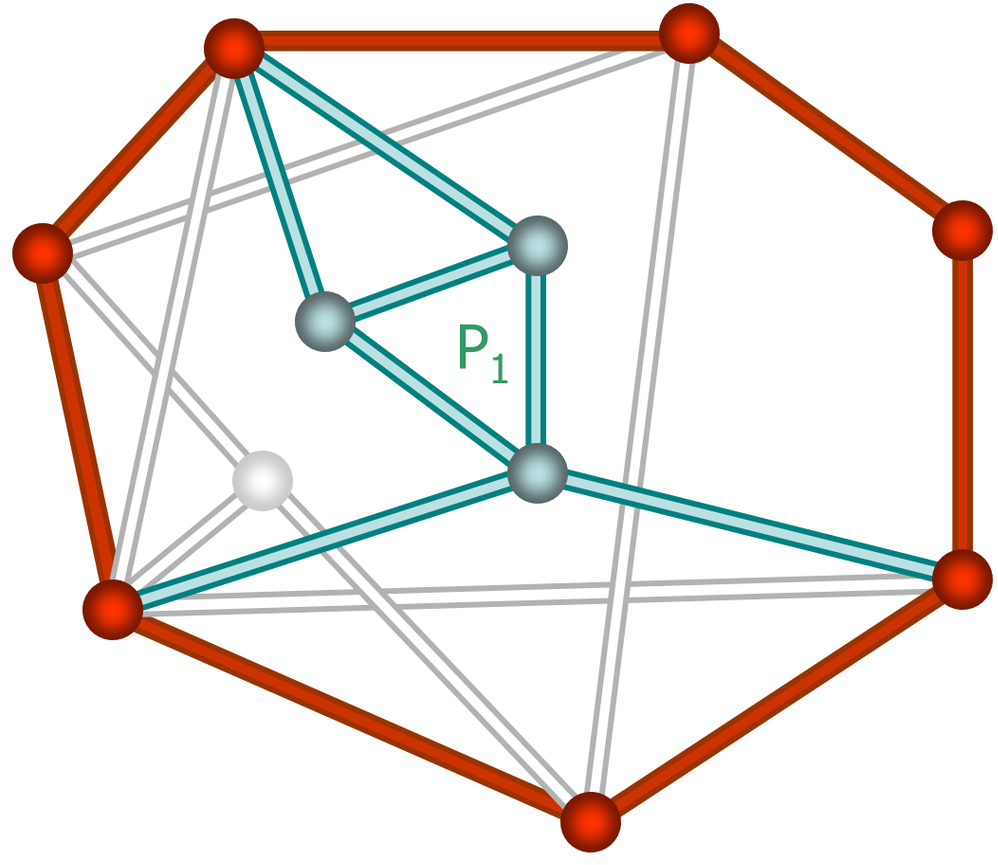
# Find a cycle that splits more than one piece

- What is a “piece”?



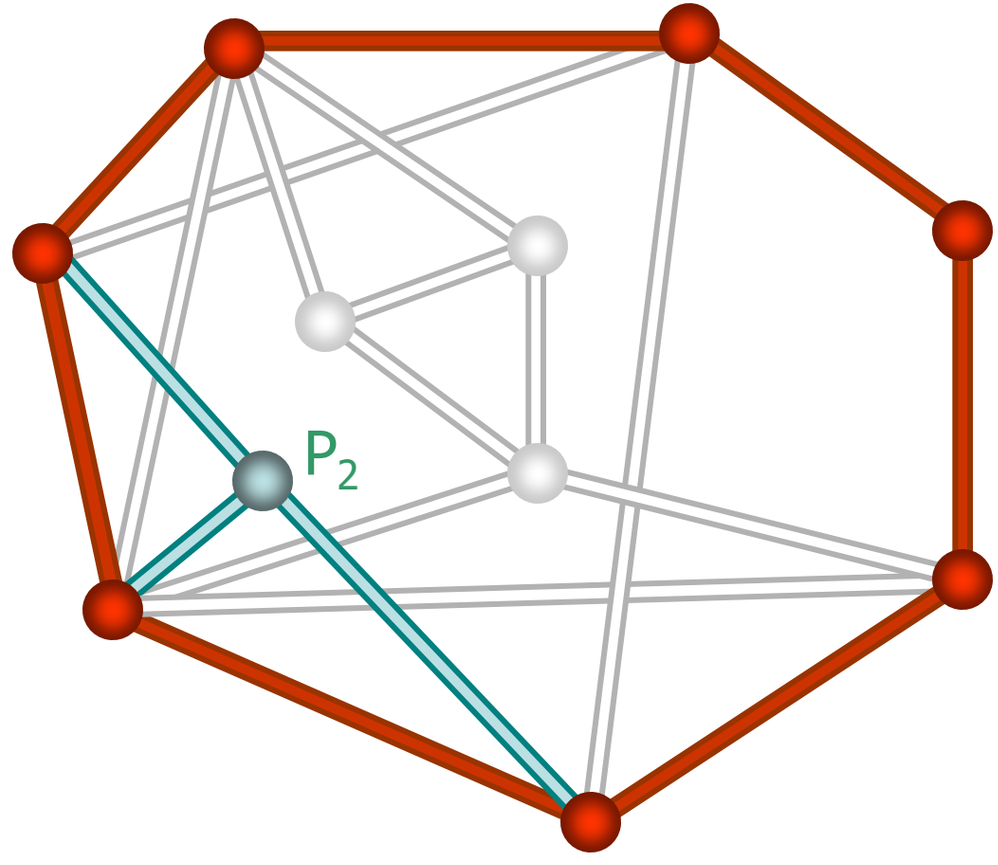

 copyright ©2025 g.di battista, m.patrignani

# First piece



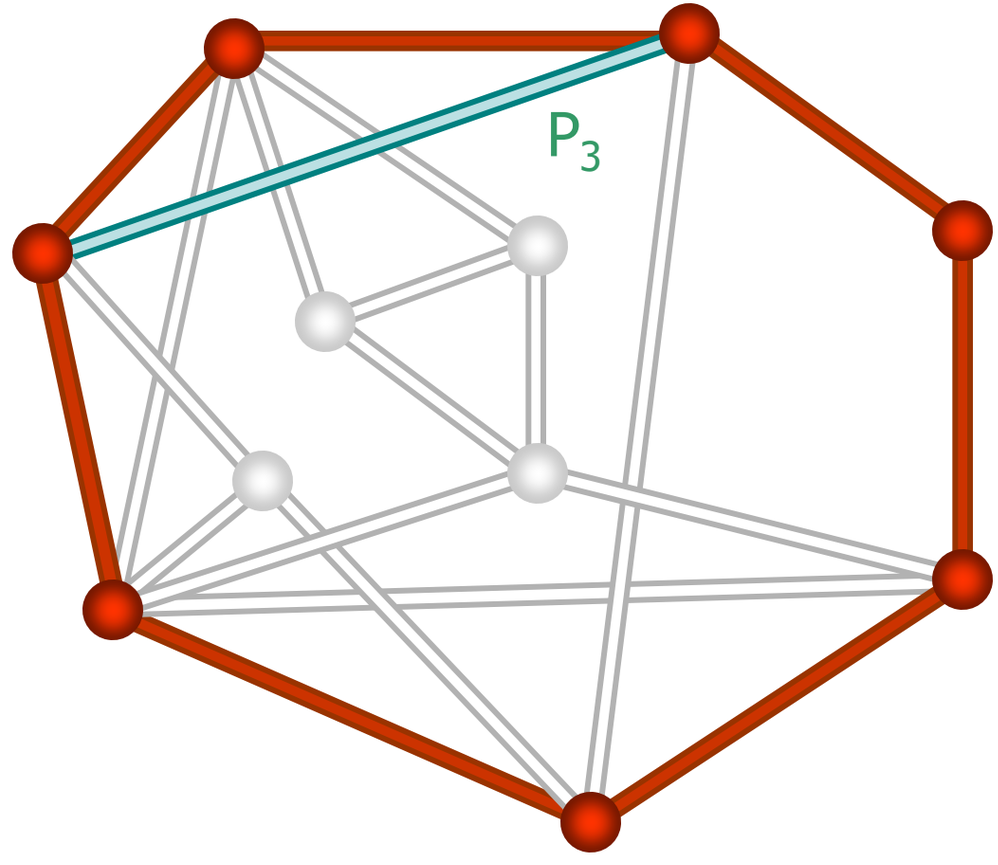

 copyright ©2025 g.di battista, m.patrignani

# Second piece



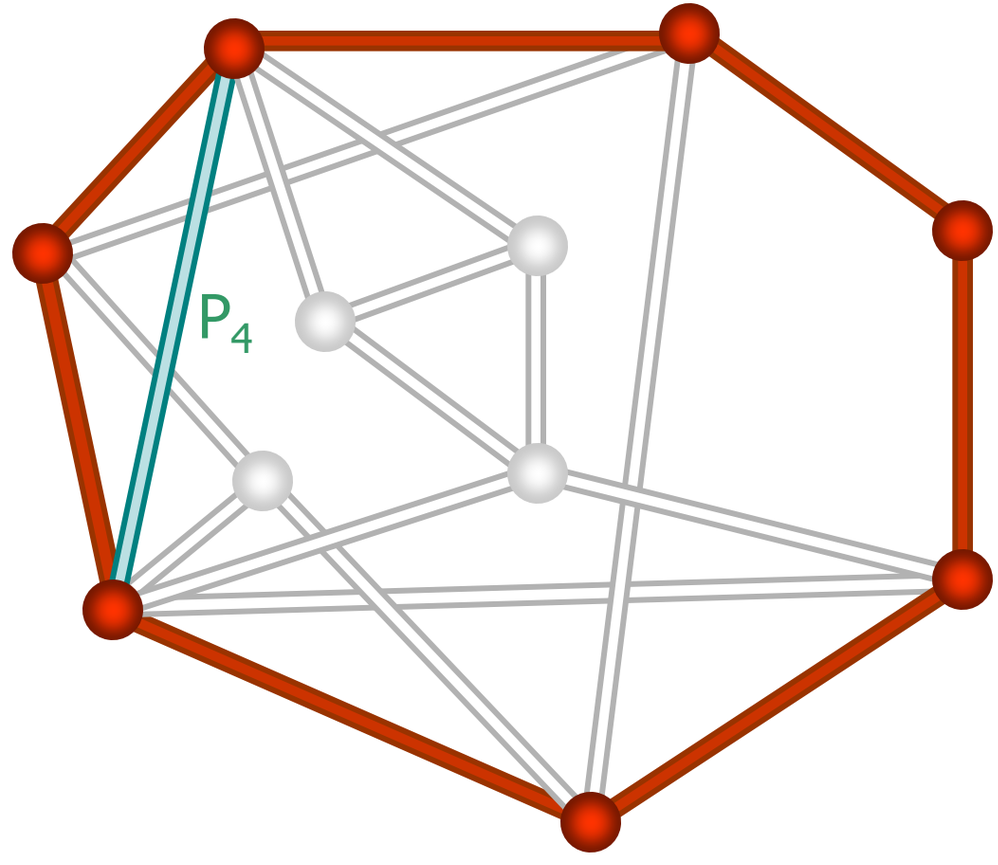

 copyright ©2025 g.di battista, m.patrignani

# Third piece



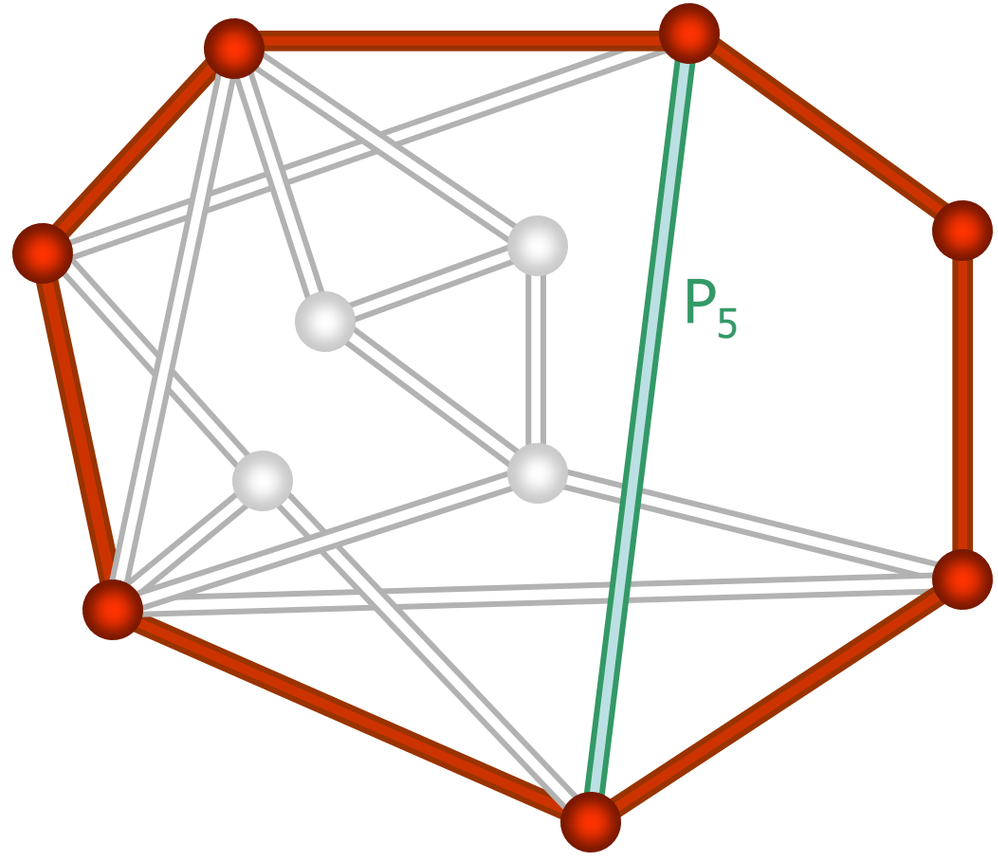

 copyright ©2025 g.di battista, m.patrignani

# Fourth piece



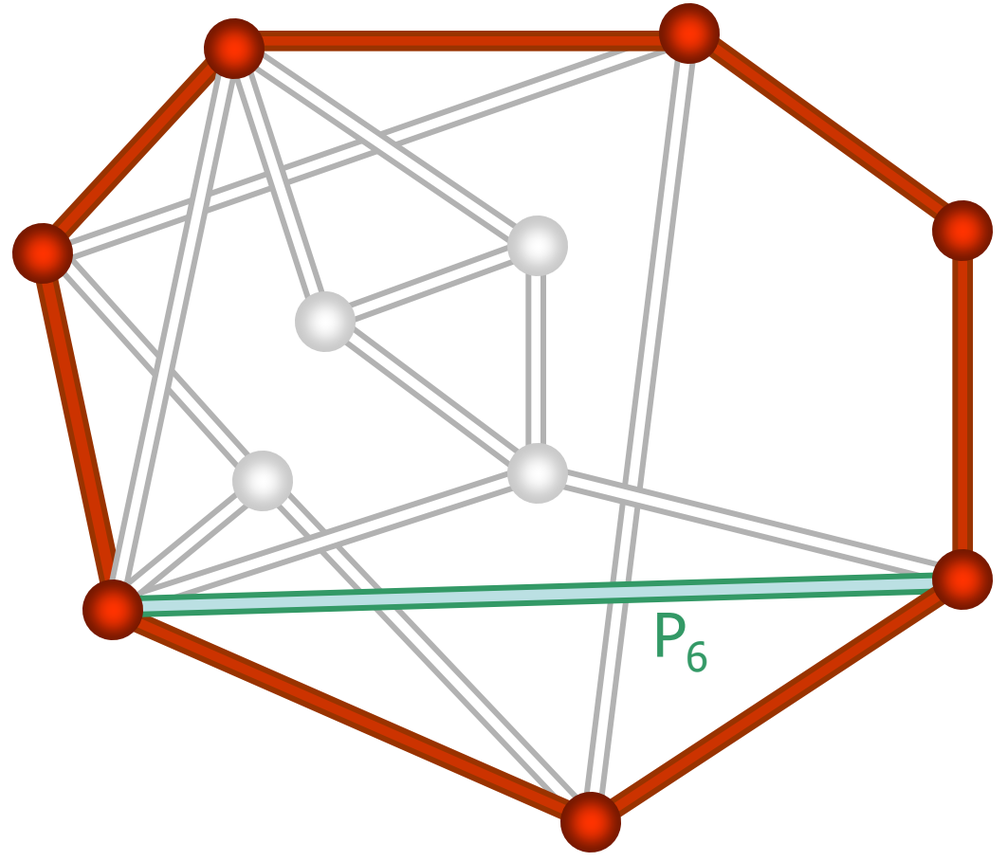

 copyright ©2025 g.di battista, m.patrignani

# Fifth piece



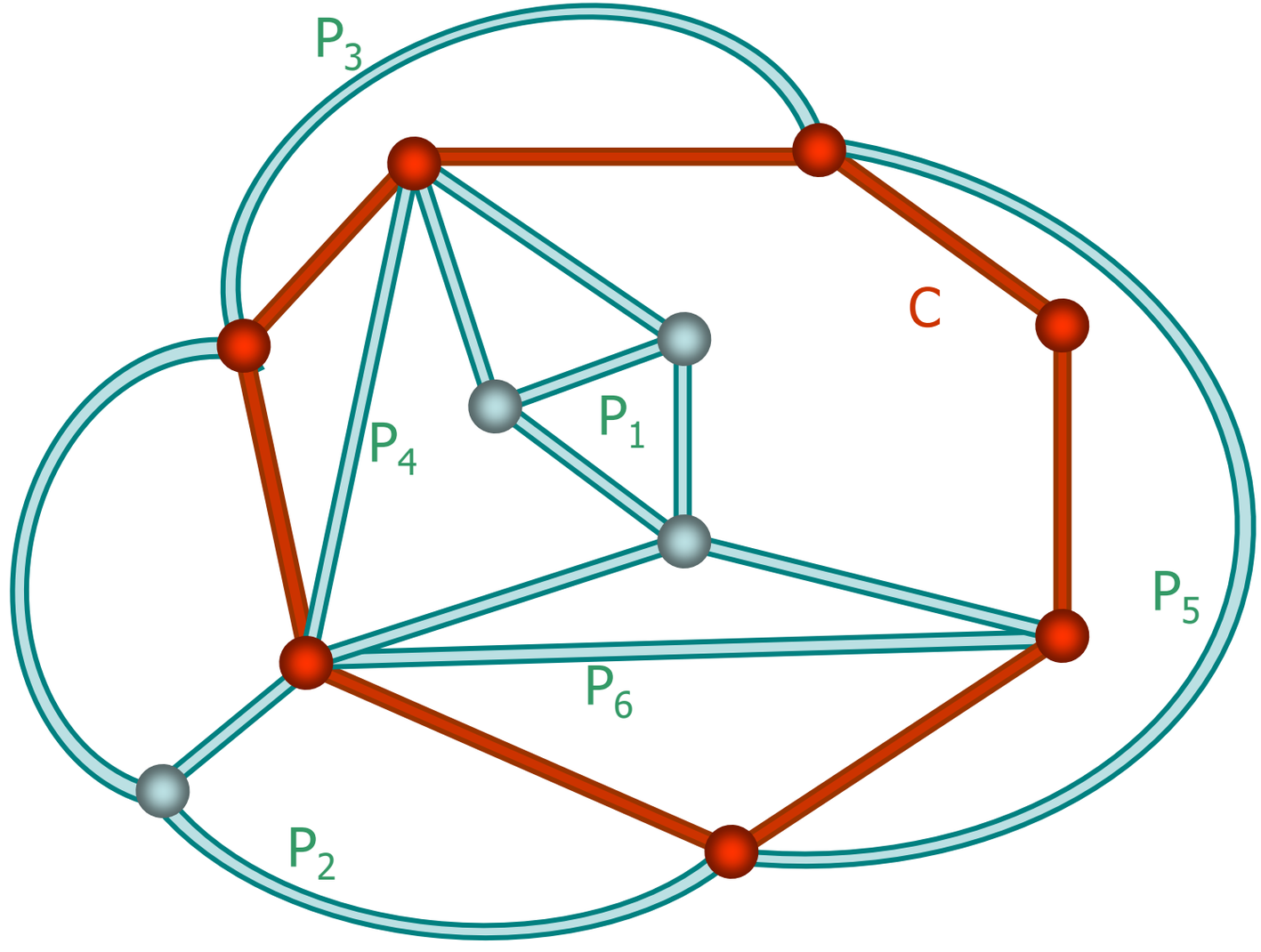

 copyright ©2025 g.di battista, m.patrignani

# Sixth piece



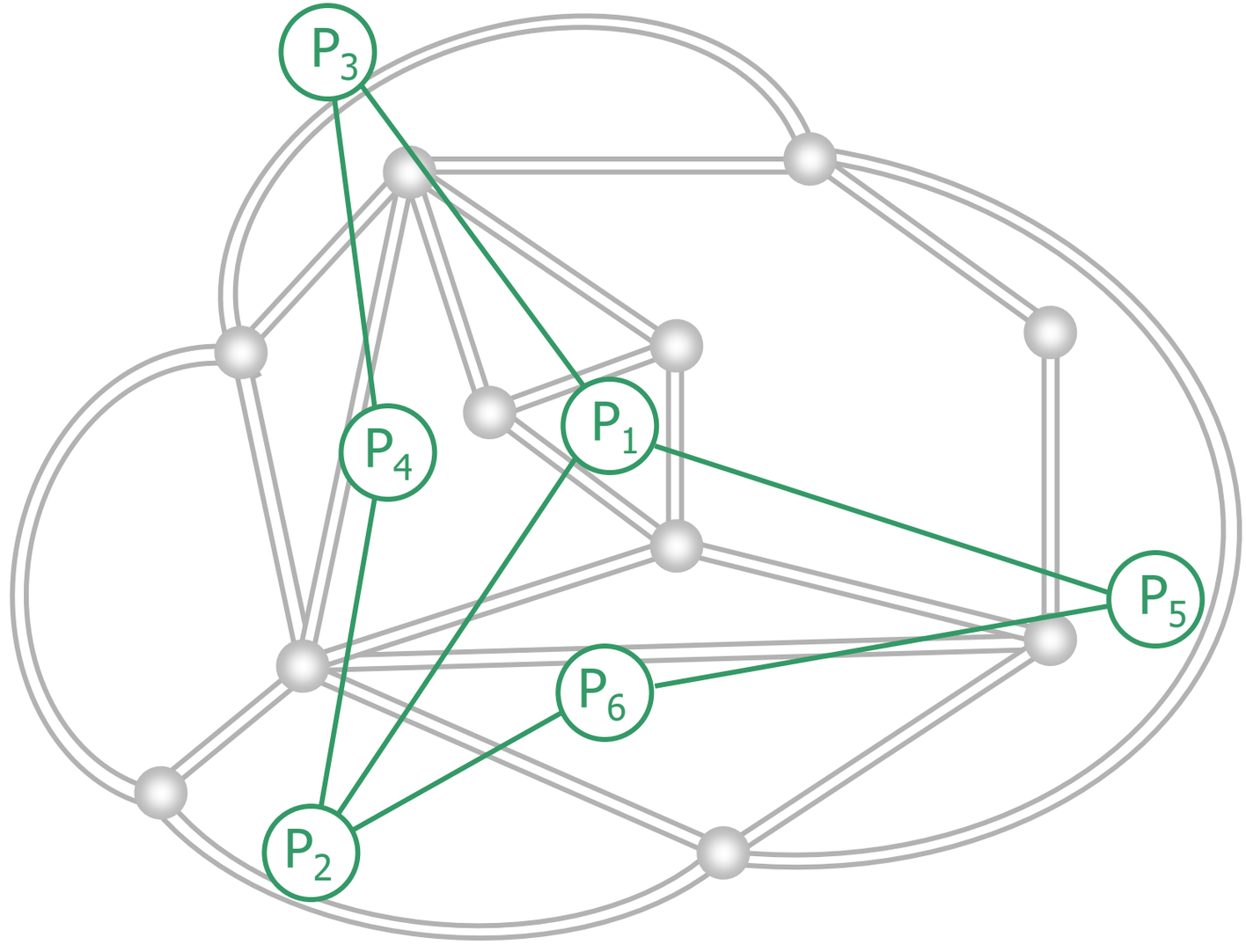

 copyright ©2025 g.di battista, m.patrignani

# All the pieces together



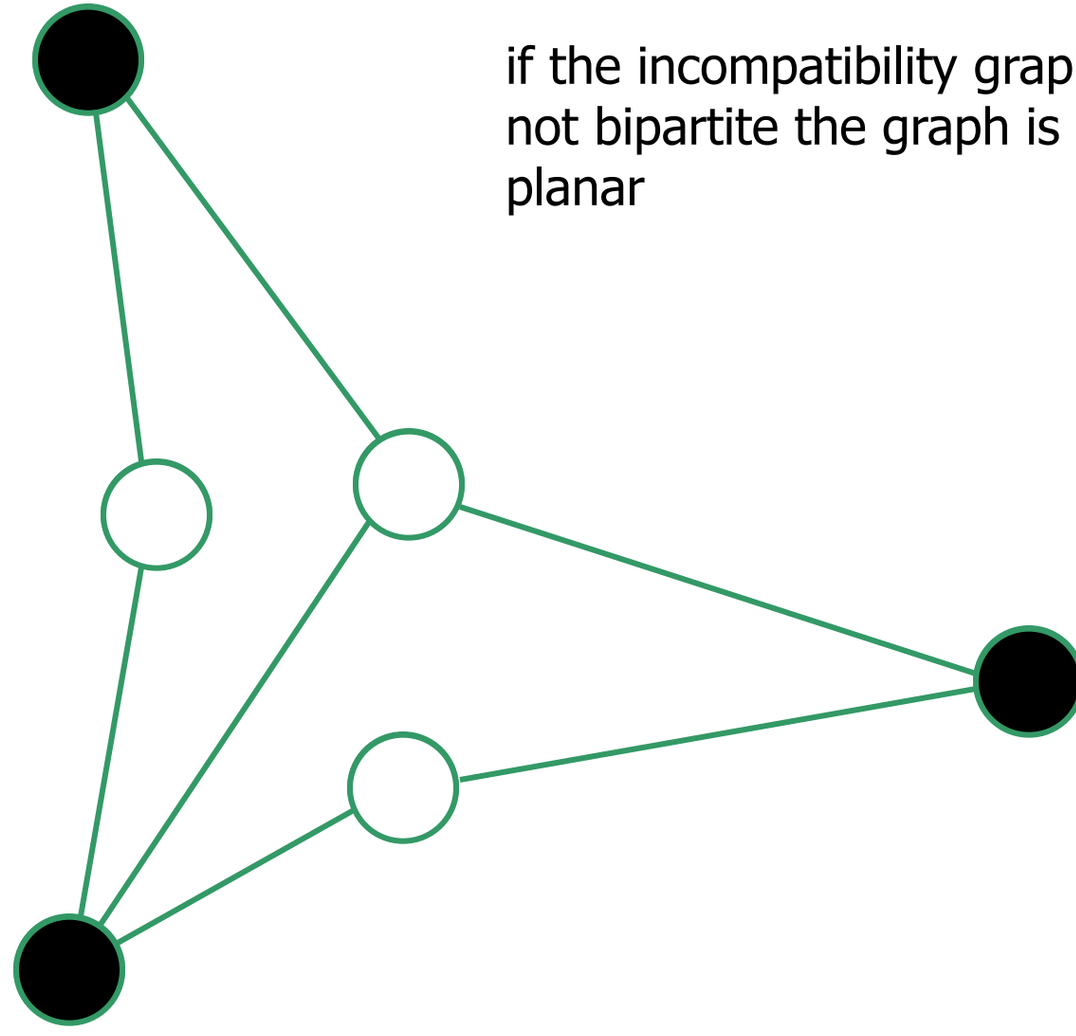

 copyright ©2025 g.di battista, m.patrignani

# Incompatibility graph




 copyright ©2025 g.di battista, m.patrignani

# Incompatibility graph

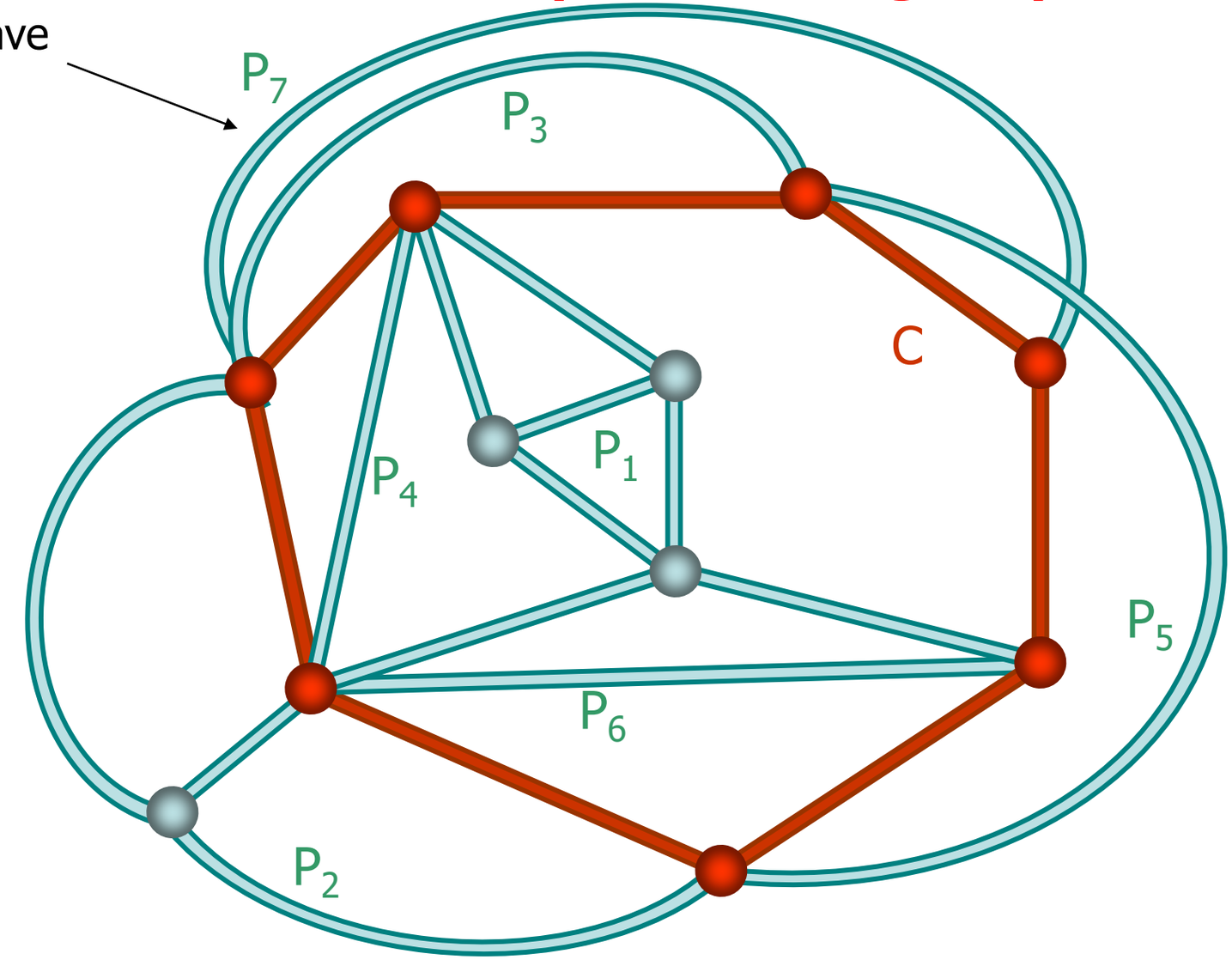


if the incompatibility graph is not bipartite the graph is not planar

 copyright ©2025 g.di battista, m.patrignani

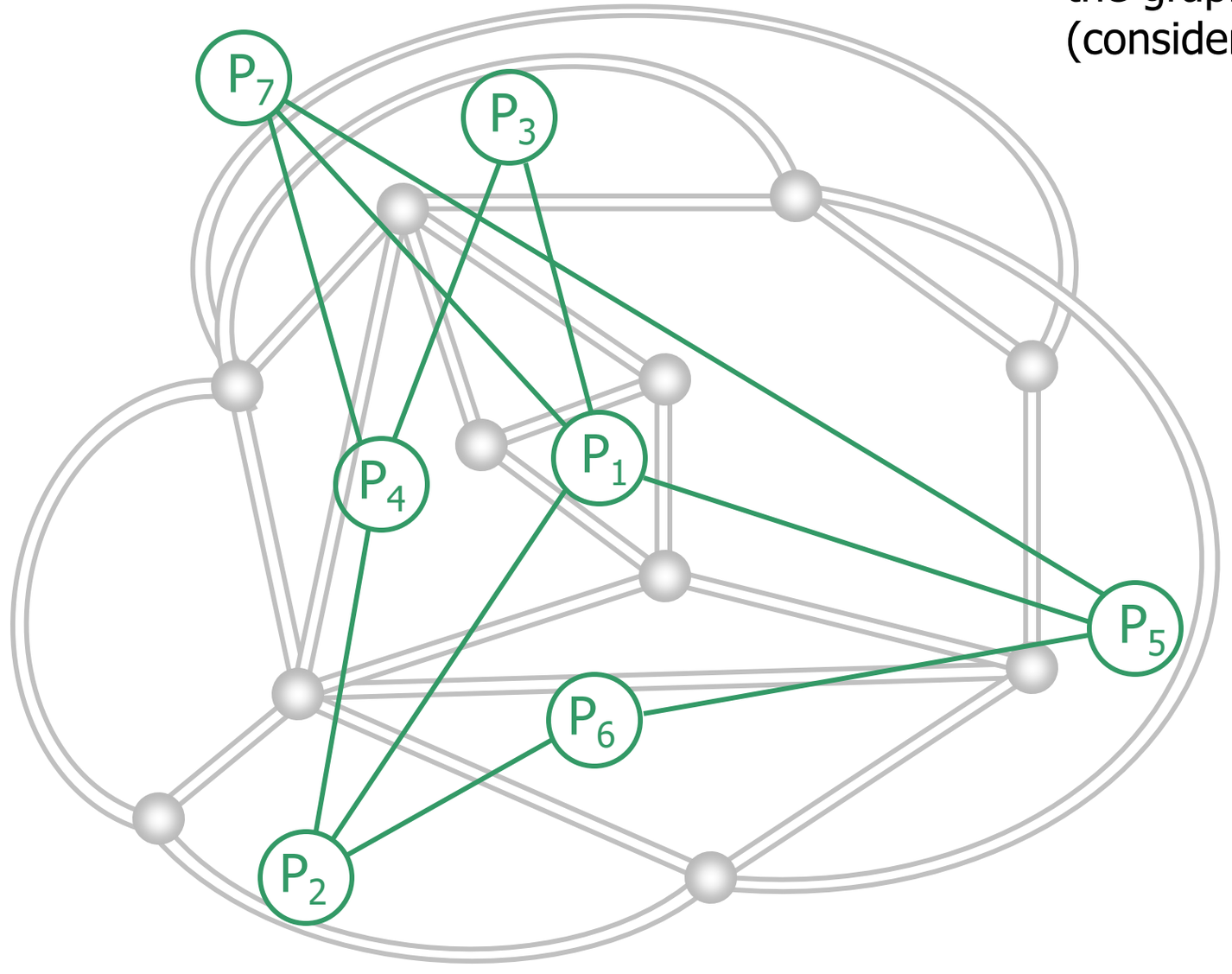
# A non-planar graph

I added $P_7$ to have a non-planar graph




 copyright ©2025 g.di battista, m.patrignani

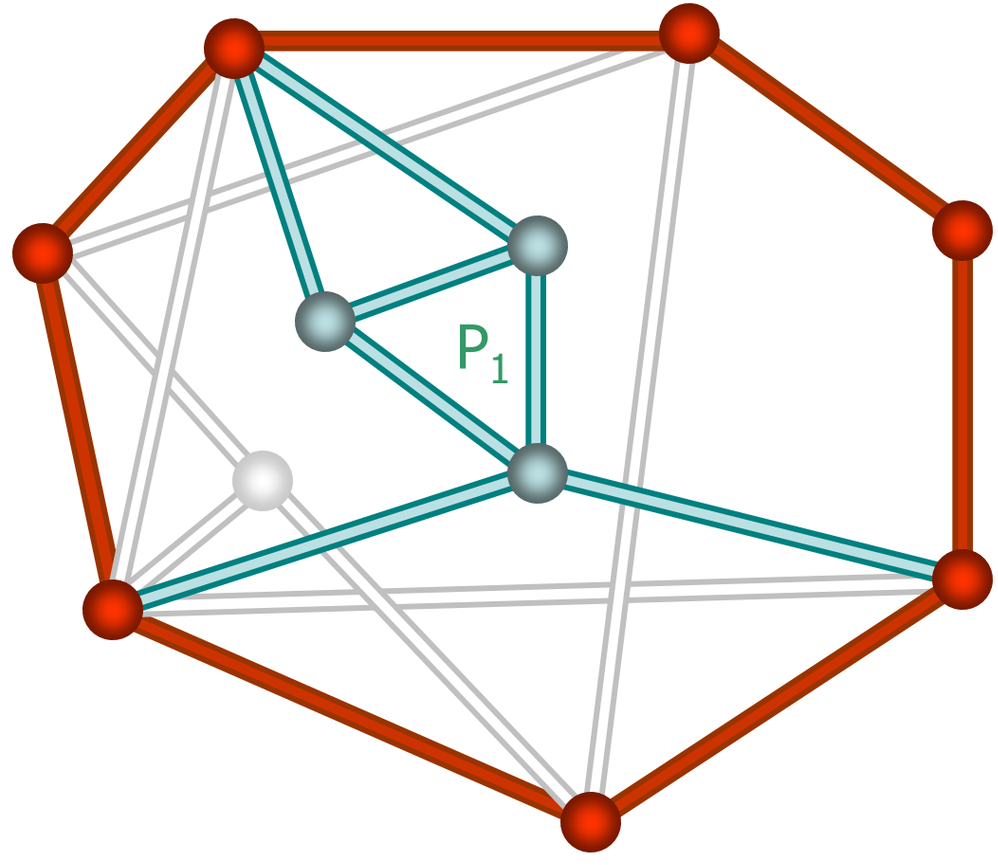
the graph is not bipartite
(consider the triangle $P_7$, $P_1$, $P_5$)




 copyright ©2025 g.di battista, m.patrignani

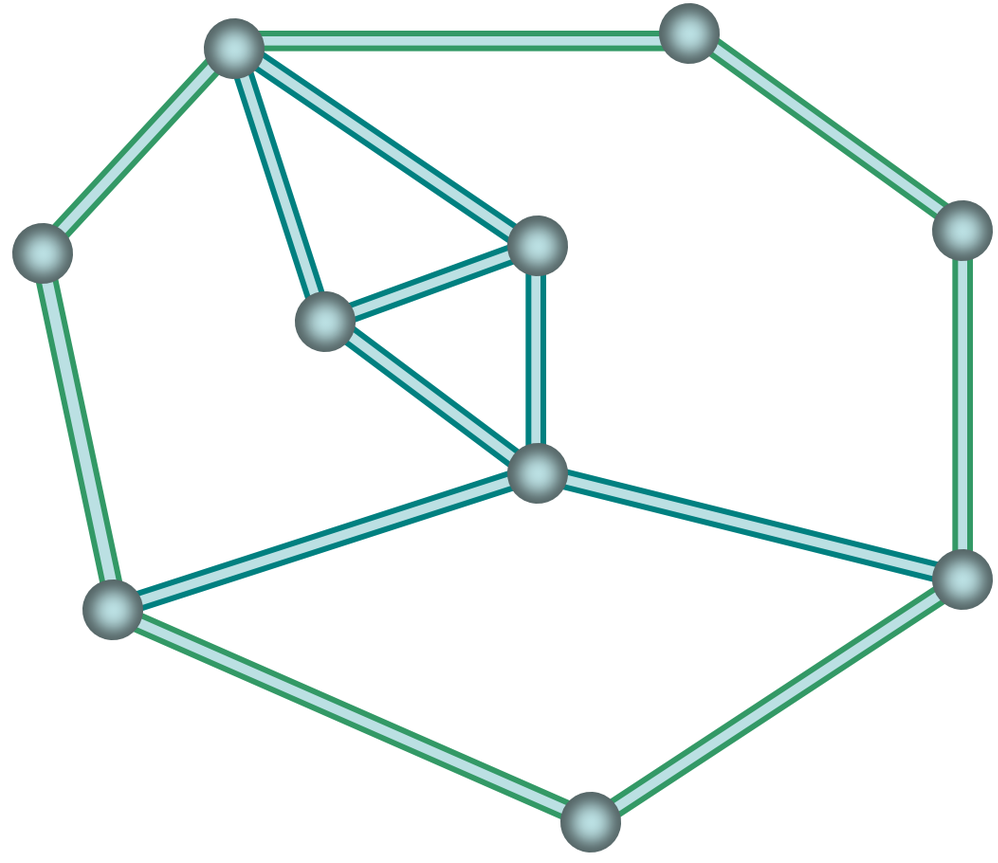
# Divide and conquer recursion

- Start again the algorithm with the cycle and each piece



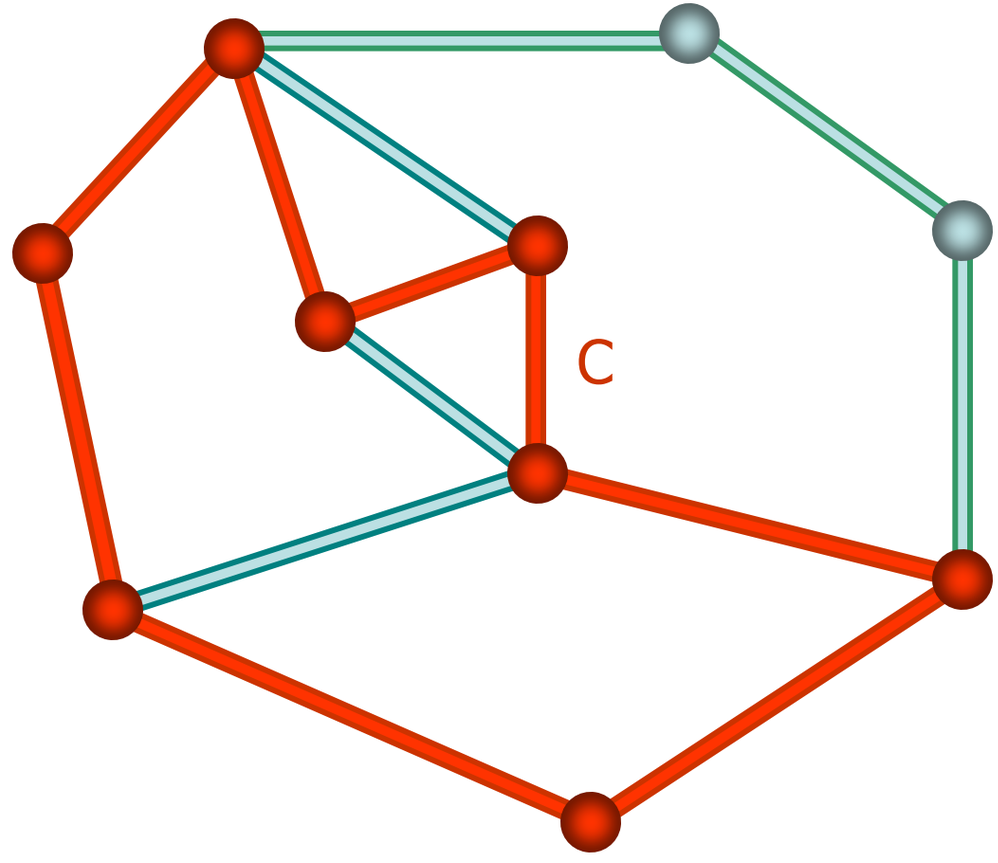

 copyright ©2025 g.di battista, m.patrignani

# Divide and conquer recursion



 copyright ©2025 g.di battista, m.patrignani

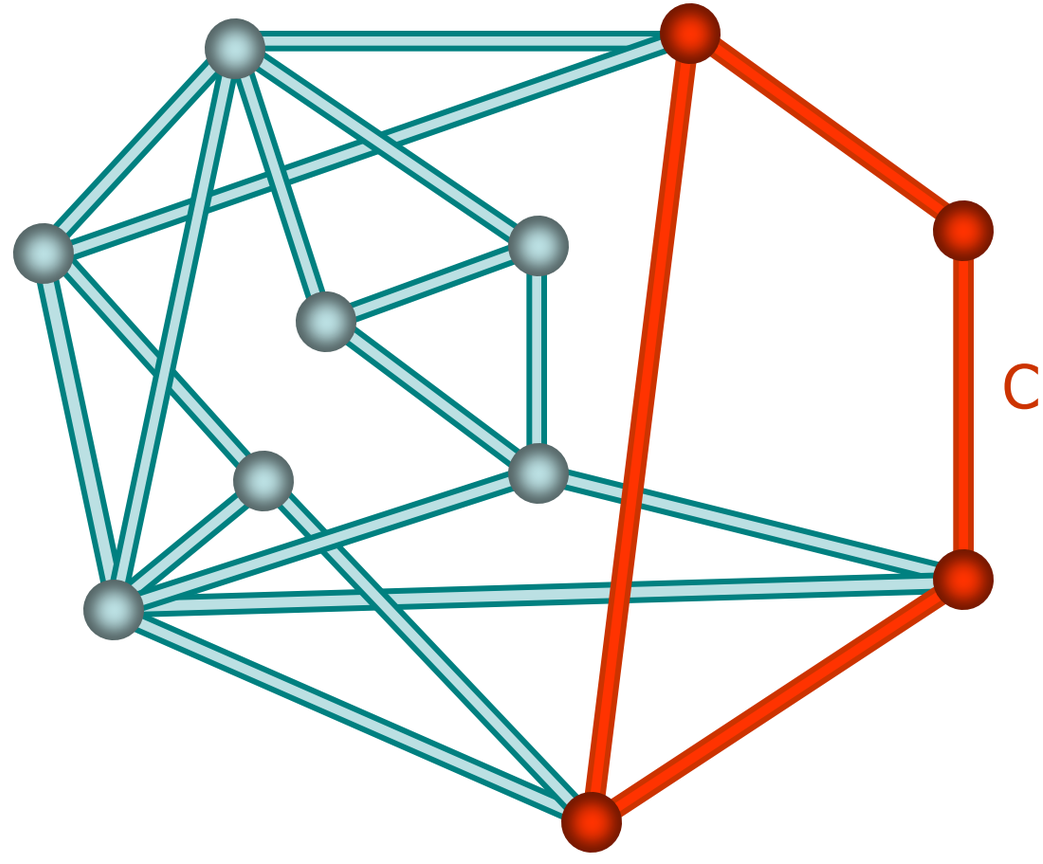
# Divide and conquer recursion




 copyright ©2025 g.di battista, m.patrignani

# Problem: you got a wrong cycle

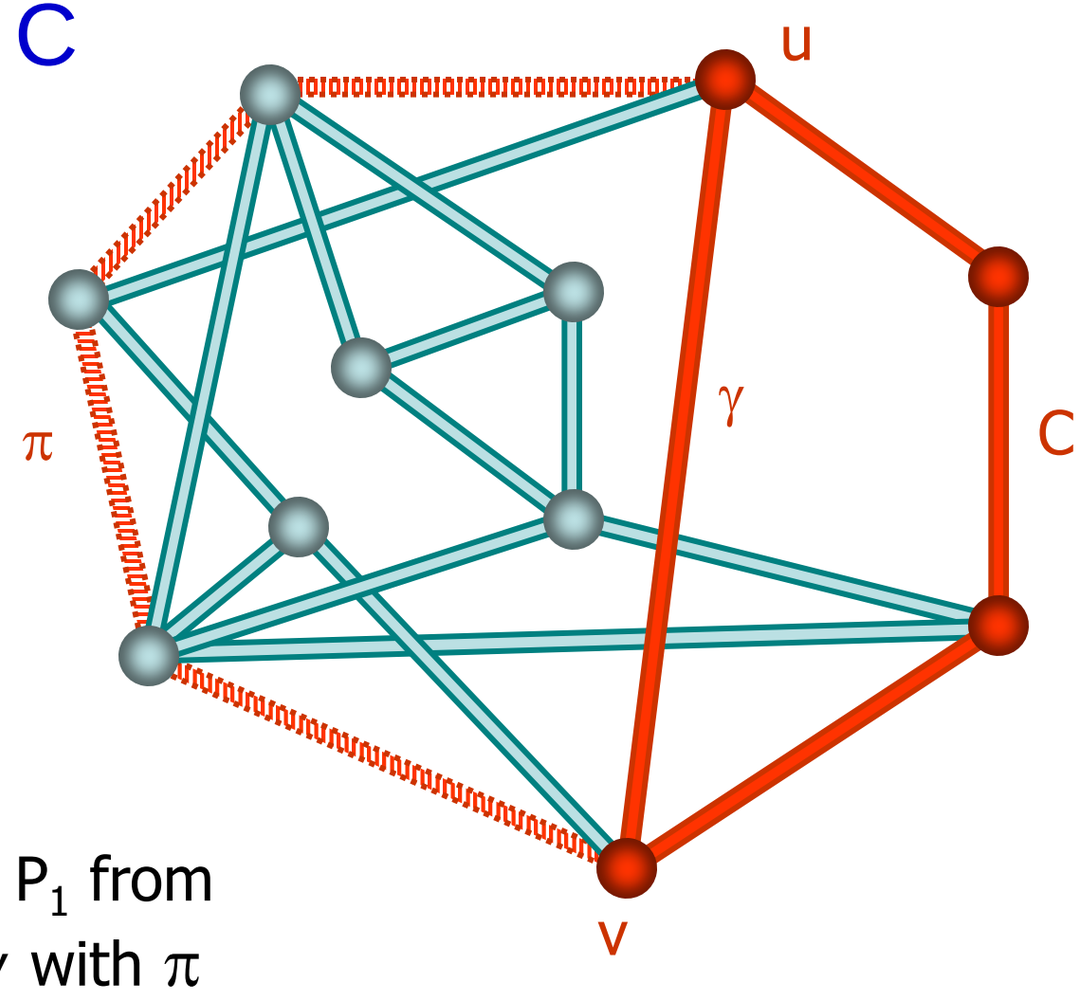
- Your cycle has a single piece




 copyright ©2025 g.di battista, m.patrignani

# You can easily change your cycle

- Take two consecutive vertices u and v where P1 attaches to C



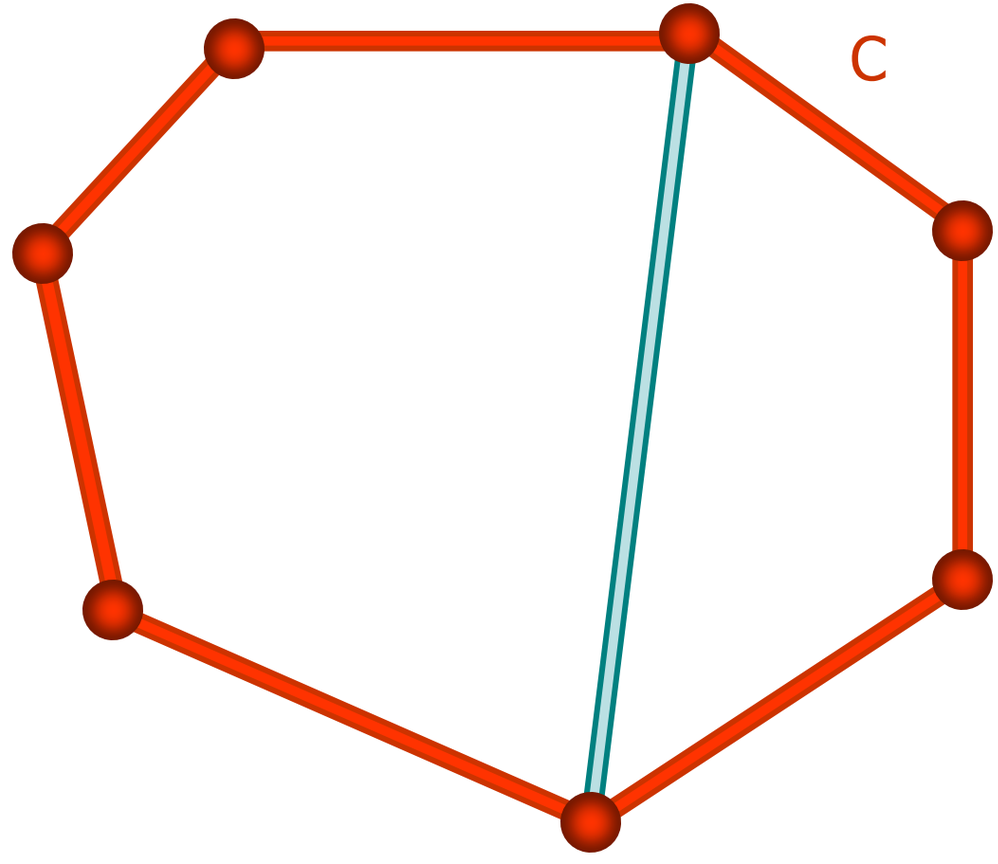

find a path $\pi$ inside $P_1$ from u to v and replace $\gamma$ with $\pi$

the new cycle has at least two pieces

 copyright ©2025 g.di battista, m.patrignani

# When the algorithm stops

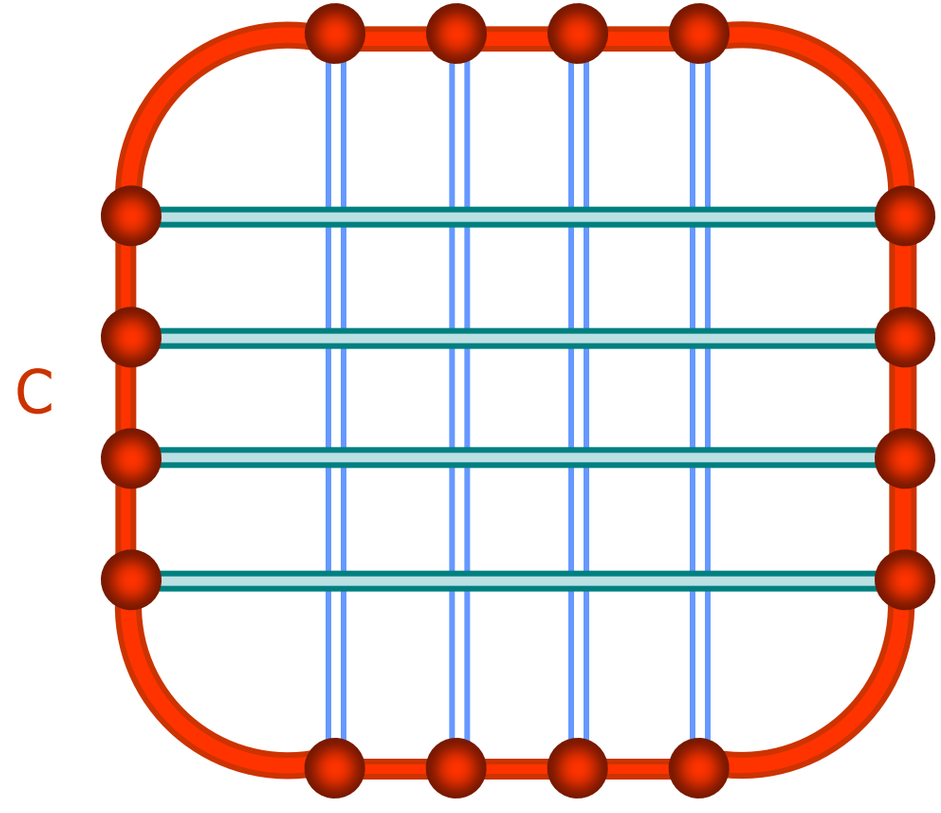
- The base case is when the only piece is a path or an edge




 copyright ©2025 g.di battista, m.patrignani

# Why complexity is at least $\Omega(n^2)$

- In certain cases the size of the incompatibility graph is quadratic
  - even if the graph is planar




 copyright ©2025 g.di battista, m.patrignani

# Why complexity is $O(n^2)$

- Observe that you could implement the Auslander and Parter algorithm as follows
  1. generate all the instances of the incompatibility graph, one for each recursion of the algorithm
  2. determine whether all such instances are bipartite with a single run of the test algorithm
     - as if it were a single non-connected instance
- This implementation would have the same complexity of the implementation where bipartiteness check is performed on each instance

 copyright ©2025 g.di battista, m.patrignani

# Why complexity is $O(n^2)$

- The number of nodes of the huge non-connected incompatibility graph is the total number of pieces in all recursions of the algorithm
- This number is the degree of the recursion tree of Auslander and Parter
- The size of the recursion tree is linear since each leave has at least an exclusive path of the input graph
- Therefore, the complexity of Auslander and Parter is $O(n^2)$ and $\Theta(n^2)$

 copyright ©2025 g.di battista, m.patrignani

# Bibliography and further readings


- [Auslander & Parter 61] L. Auslander and S. V. Parter. “On imbedding graphs in the sphere”. Journal of Mathematics and Mechanics, 1961
- [Di Battista *et al.* 99] G. Di Battista, P. Eades, R. Tamassia, and I. G. Tollis. Graph Drawing. Prentice Hall, Upper Saddle River, NJ, 1999
- [Hopcroft & Tarjan 74] J. Hopcroft and R. E. Tarjan. Efficient planarity testing. J. ACM, 1974
- [Mehlhorn & Mutzel 96] K. Mehlhorn and P. Mutzel. On the embedding phase of the Hopcroft and Tarjan planarity testing algorithm. Algorithmica, 1996


 copyright ©2025 g.di battista, m.patrignani